



US012405568B2

(12) **United States Patent**
Okabe et al.

(10) **Patent No.:** **US 12,405,568 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **IMAGE FORMING APPARATUS INCLUDING COUPLING MEMBER SELECTIVELY COUPLED TO PHOTSENSITIVE DRUM**

(71) Applicant: **BROTHER KOGYO KABUSHIKI KAISHA**, Nagoya (JP)

(72) Inventors: **Yasushi Okabe**, Nagoya (JP); **Junichi Hashimoto**, Toyohashi (JP)

(73) Assignee: **BROTHER KOGYO KABUSHIKI KAISHA**, Nagoya (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/647,889**

(22) Filed: **Apr. 26, 2024**

(65) **Prior Publication Data**
US 2024/0345537 A1 Oct. 17, 2024

Related U.S. Application Data
(60) Continuation of application No. 17/752,219, filed on May 24, 2022, now Pat. No. 12,007,714, which is a (Continued)

(30) **Foreign Application Priority Data**
Oct. 31, 2008 (JP) 2008-281991
Oct. 31, 2008 (JP) 2008-281992
Oct. 31, 2008 (JP) 2008-281993

(51) **Int. Cl.**
G03G 21/18 (2006.01)
G03G 15/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G03G 21/1853** (2013.01); **G03G 15/0194** (2013.01); **G03G 15/5008** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G03G 15/0194; G03G 15/5008; G03G 15/757; G03G 21/1671; G03G 21/1857; G03G 21/186; G03G 21/1871
See application file for complete search history.

(56) **References Cited**
U.S. PATENT DOCUMENTS
3,638,112 A 1/1972 Lasater et al.
5,765,082 A 6/1998 Numazu et al.
(Continued)

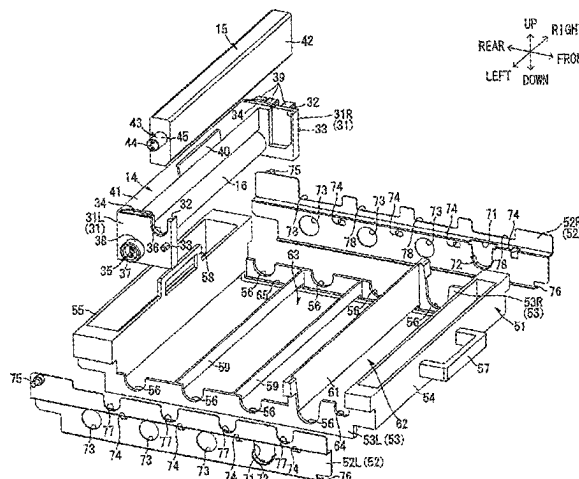
FOREIGN PATENT DOCUMENTS
EP 0058161 A1 12/2000
EP 1816527 A2 8/2007
(Continued)

OTHER PUBLICATIONS
Office Action received in corresponding Chinese Patent Application 200910205932.9 mailed Jun. 1, 2012.
(Continued)

Primary Examiner — Benjamin R Schmitt
(74) *Attorney, Agent, or Firm* — Merchant & Gould P.C.

(57) **ABSTRACT**
In an image forming apparatus, a process frame is movable into and out of a main casing in a first direction. The process frame includes a first wall and a second wall spaced apart from the first wall in a second direction. A first cartridge is detachably mountable to the process frame in a third direction crossing the second direction. A first electrode protrudes from an outer surface of the first cartridge outwardly in the second direction. In a state where the first cartridge is mounted to the process frame, the first cartridge is positioned between the first wall and second wall of the process frame in the second direction and a portion of the first wall aligns with the first electrode in the third direction such that the portion of the first wall is overlapped with the first electrode as viewed in the third direction.

14 Claims, 22 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/089,879, filed on Nov. 5, 2020, now Pat. No. 11,353,823, which is a continuation of application No. 16/573,094, filed on Sep. 17, 2019, now Pat. No. 10,831,152, which is a continuation of application No. 16/130,438, filed on Sep. 13, 2018, now Pat. No. 10,444,700, which is a continuation of application No. 15/275,905, filed on Sep. 26, 2016, now Pat. No. 10,082,766, which is a continuation of application No. 14/948,591, filed on Nov. 23, 2015, now Pat. No. 9,477,202, which is a continuation of application No. 14/053,143, filed on Oct. 14, 2013, now Pat. No. 9,201,370, which is a division of application No. 12/606,660, filed on Oct. 27, 2009, now Pat. No. 8,577,255.

(51) **Int. Cl.****G03G 15/01** (2006.01)**G03G 21/16** (2006.01)(52) **U.S. Cl.**

CPC **G03G 15/757** (2013.01); **G03G 21/1647** (2013.01); **G03G 21/1652** (2013.01); **G03G 21/1825** (2013.01); **G03G 21/1842** (2013.01); **G03G 21/1867** (2013.01); **G03G 15/0178** (2013.01); **G03G 2221/166** (2013.01); **G03G 2221/1684** (2013.01); **G03G 2221/1869** (2013.01)

(56)

References Cited**U.S. PATENT DOCUMENTS**

6,708,011	B2	3/2004	Nomura et al.
6,721,521	B2	4/2004	Kataoka et al.
7,706,716	B2	4/2010	Tada et al.
7,715,752	B2	5/2010	Sakurai et al.
7,826,773	B2	11/2010	Kawamura
7,899,360	B2	3/2011	Kondo et al.
7,937,020	B2	5/2011	Sakurai et al.
7,957,670	B2	6/2011	Kondo et al.
8,056,894	B2	11/2011	Ichikawa
8,126,366	B2	2/2012	Hattori
9,116,502	B2	8/2015	Hayakawa et al.
9,367,026	B2	6/2016	Nakashima
2001/0019418	A1	9/2001	Kataoka et al.
2003/0161656	A1	8/2003	Miura et al.
2005/0214023	A1	9/2005	Foster et al.
2006/0140674	A1	6/2006	Sato
2006/0146371	A1	7/2006	Hoashi et al.
2007/0110473	A1	5/2007	Kondo et al.
2007/0160384	A1	7/2007	Sakurai et al.
2007/0160385	A1	7/2007	Noguchi et al.
2007/0177899	A1	8/2007	Kawamura
2007/0183814	A1	8/2007	Kamimura
2007/0280730	A1	12/2007	Okabe et al.
2007/0280733	A1	12/2007	Shiraki et al.
2007/0286632	A1	12/2007	Okabe
2008/0002341	A1	1/2008	Tomatsu
2008/0003015	A1	1/2008	Tomatsu
2008/0075502	A1	3/2008	Tada et al.
2008/0159772	A1	7/2008	Koishi et al.
2008/0159774	A1	7/2008	Tanabe et al.
2008/0159781	A1	7/2008	Noguchi et al.
2008/0310881	A1	12/2008	Sato
2009/0041498	A1	2/2009	Numagami et al.
2009/0047040	A1	2/2009	Nieda
2009/0123181	A1	5/2009	Ito et al.
2009/0169247	A1	7/2009	Fattori
2009/0202272	A1	8/2009	Nakashima
2009/0245869	A1	10/2009	Nieda
2010/0111562	A1	5/2010	Okabe et al.
2010/0209139	A1	8/2010	Sakurai et al.

2011/0013935	A1	1/2011	Kondo et al.
2011/0102983	A1	5/2011	Souda et al.
2011/0164897	A1	7/2011	Kikuchi
2012/0114376	A1	5/2012	Kawamura
2012/0195633	A1	8/2012	Okabe
2012/0251172	A1	10/2012	Hashimoto et al.
2012/0275822	A1	11/2012	Kawamura
2014/0010569	A1	1/2014	Souda
2014/0086622	A1	3/2014	Nakashima
2014/0093279	A1	4/2014	Hayakawa et al.
2014/0241749	A1	8/2014	Okabe
2014/0294437	A1	10/2014	Okabe
2014/0376944	A1	12/2014	Souda et al.
2014/0376945	A1	12/2014	Souda et al.
2014/0376946	A1	12/2014	Souda et al.
2014/0376947	A1	12/2014	Nishimura
2014/0376957	A1	12/2014	Souda et al.
2014/0376958	A1	12/2014	Souda et al.
2015/0139679	A1	5/2015	Souda
2015/0139687	A1	5/2015	Souda
2015/0177677	A1	6/2015	Souda
2015/0177678	A1	6/2015	Souda
2015/0177679	A1	6/2015	Souda
2015/0177680	A1	6/2015	Souda et al.
2015/0261174	A1	9/2015	Souda
2016/0009510	A1	1/2016	Kawai

FOREIGN PATENT DOCUMENTS

JP	08-305184	A	11/1996
JP	09-281770	A	10/1997
JP	11-084799	A	3/1999
JP	2000-181329	A	6/2000
JP	2001-249517	A	9/2001
JP	2003-015378	A	1/2003
JP	2003-316233	A	11/2003
JP	2006-184901	A	7/2006
JP	2007-140041	A	6/2007
JP	2007-213023	A	8/2007
JP	2007-213033	A	8/2007
JP	2008-065122	A	3/2008
JP	2009-162915	A	7/2009
JP	2010-052423	A	3/2010
JP	2010-107890	A	5/2010
JP	2011-065030	A	3/2011
JP	2011-215406	A	10/2011
JP	4867955	B2	2/2012

OTHER PUBLICATIONS

Office Action received in corresponding Chinese Patent Application 200910205932.9 mailed May 25, 2011.

Extended European Search Report for application No. 09174172.8 mailed Nov. 24, 2010.

Office Action received in corresponding Japanese Patent Application No. 2008-281992 mailed Sep. 7, 2010.

Office Action received in corresponding Japanese Patent Application No. 2008-281993 mailed Sep. 7, 2010.

Office Action received in corresponding Japanese Patent Application 2008-281991 mailed Aug. 17, 2010.

Copending U.S. Appl. No. 14/032,988, filed Sep. 20, 2013.

Copending U.S. Appl. No. 14/032,923, filed Sep. 20, 2013.

Non-final Office Action issued in U.S. Appl. No. 14/053,095 mailed Jan. 27, 2014.

Final Office Action received in corresponding U.S. Appl. No. 14/053,095 mailed Jun. 16, 2014.

Dec. 29, 2014—(US) Non-final Office Action—U.S. Appl. No. 14/032,988.

Feb. 10, 2015—(US) Notice of Allowance—U.S. Appl. No. 14/053,095.

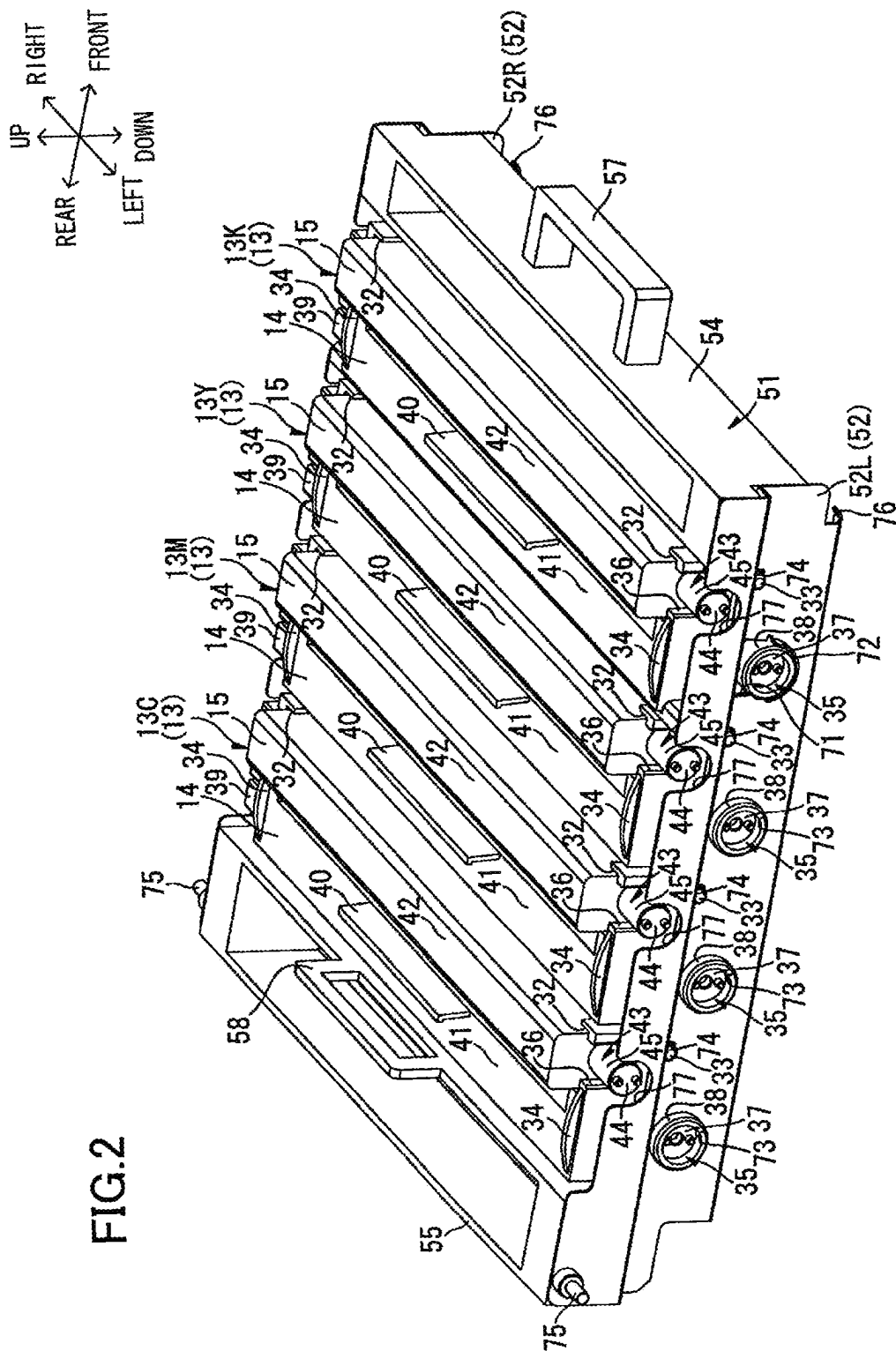
Apr. 14, 2015—(US) Non-Final Office Action—U.S. Appl. No. 14/032,923.

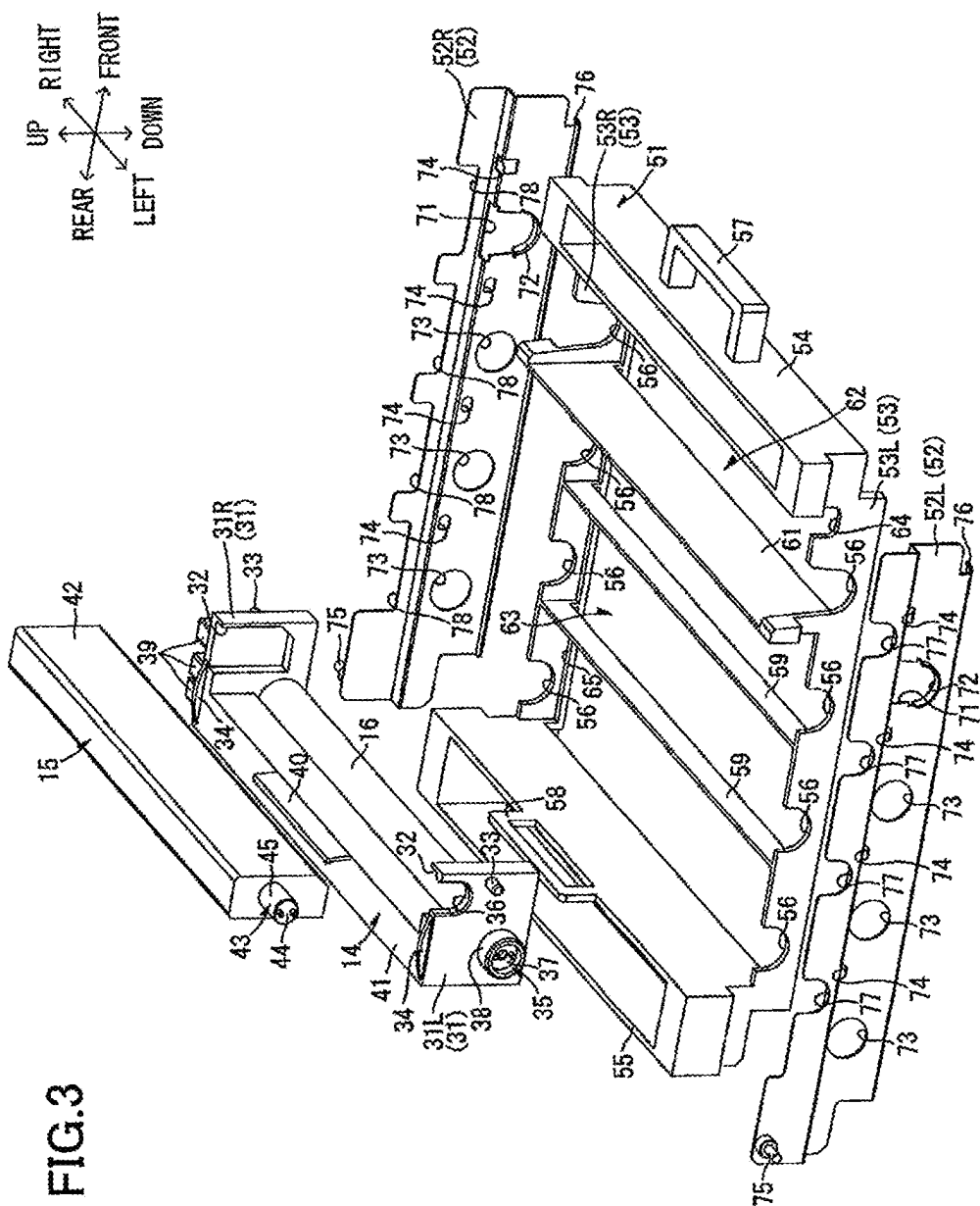
May 8, 2015—(US) Notice of Allowance—U.S. Appl. No. 14/032,988.

Oct. 9, 2015—(US) Final Office Action—U.S. Appl. No. 14/032,923.

Feb. 2, 2016—(US) Notice of Allowance—U.S. Appl. No. 14/032,923.

Apr. 6, 2016—(CN) Office Action—App 201310428796.6—Eng Tran.





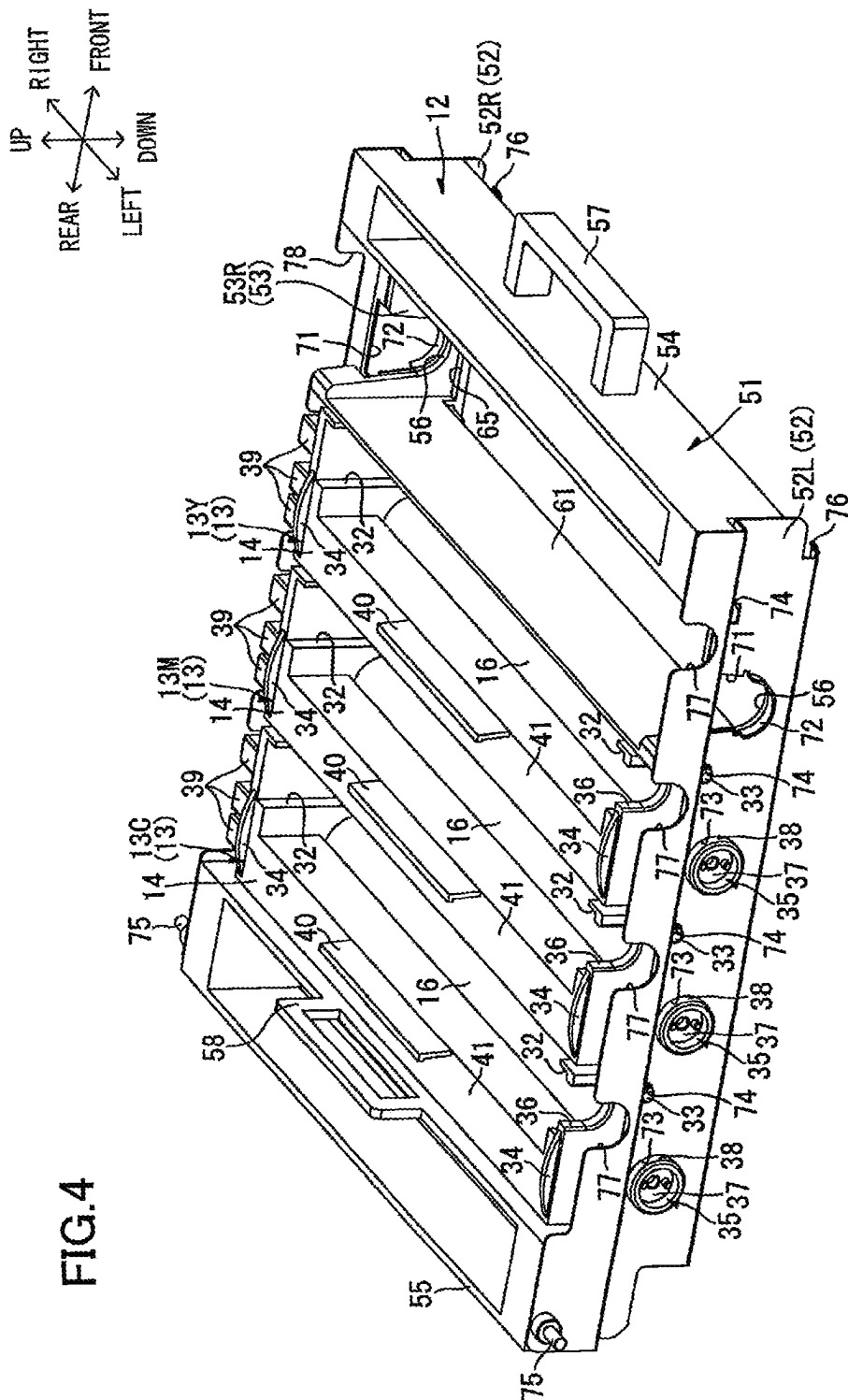
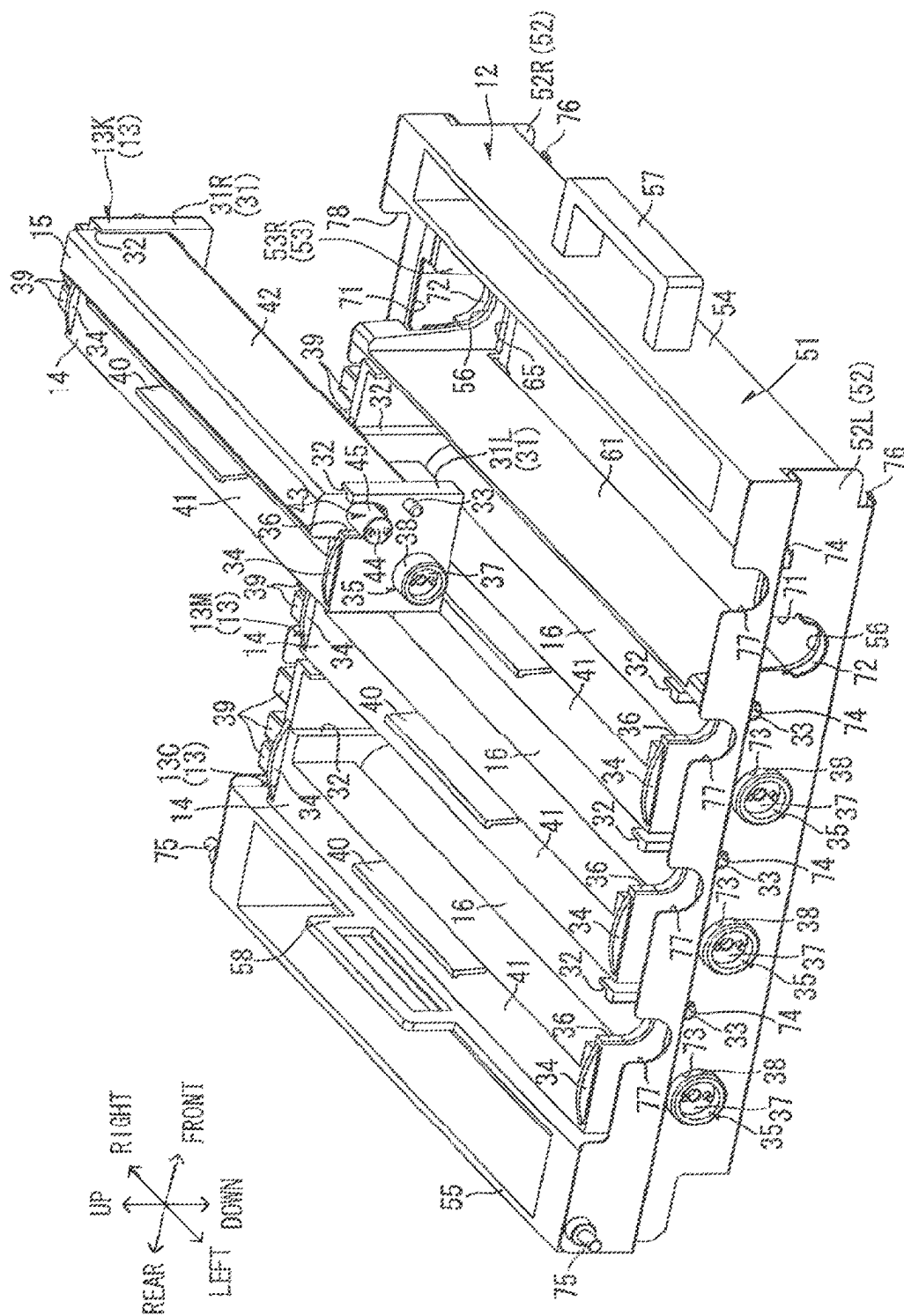
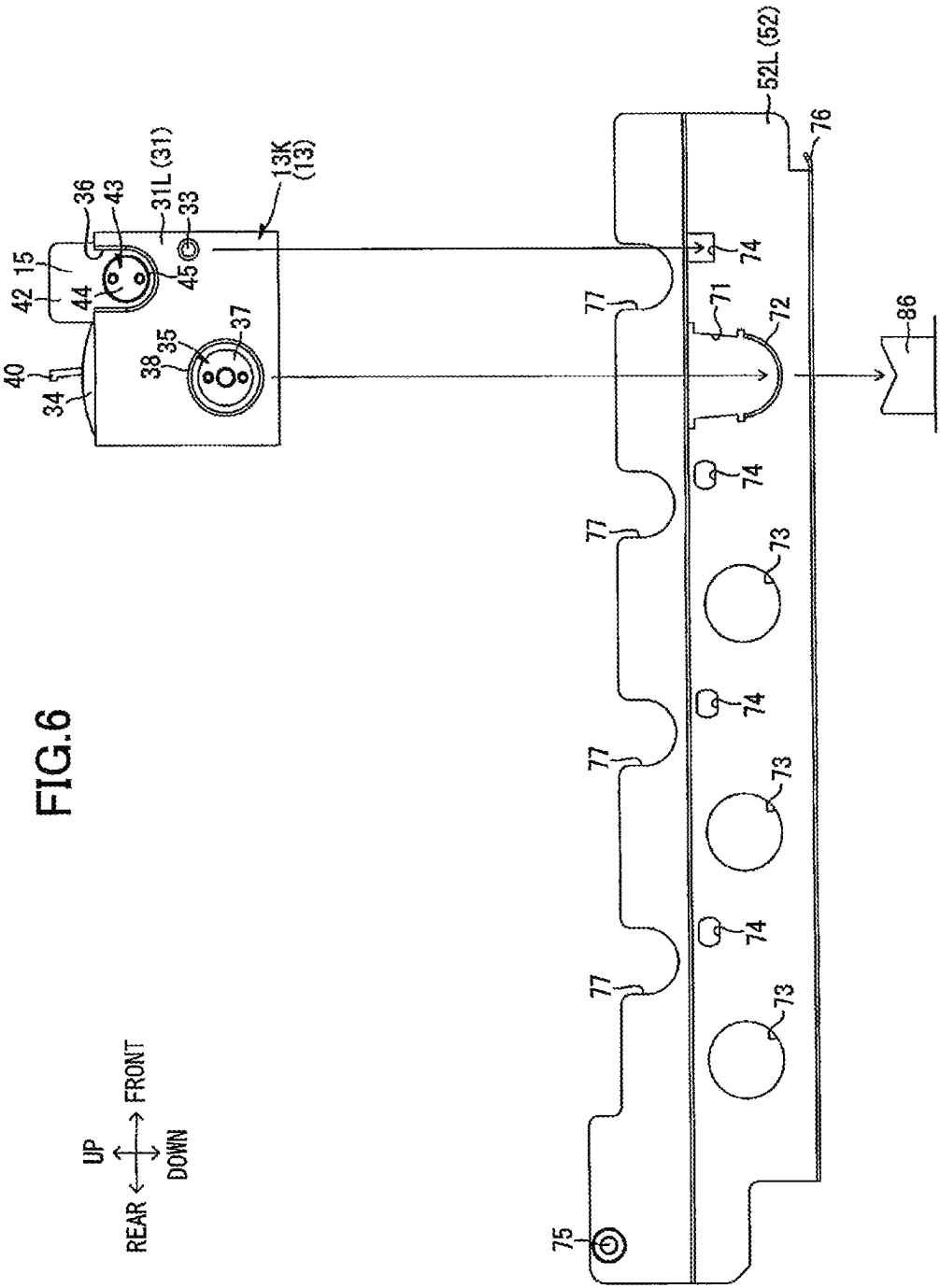
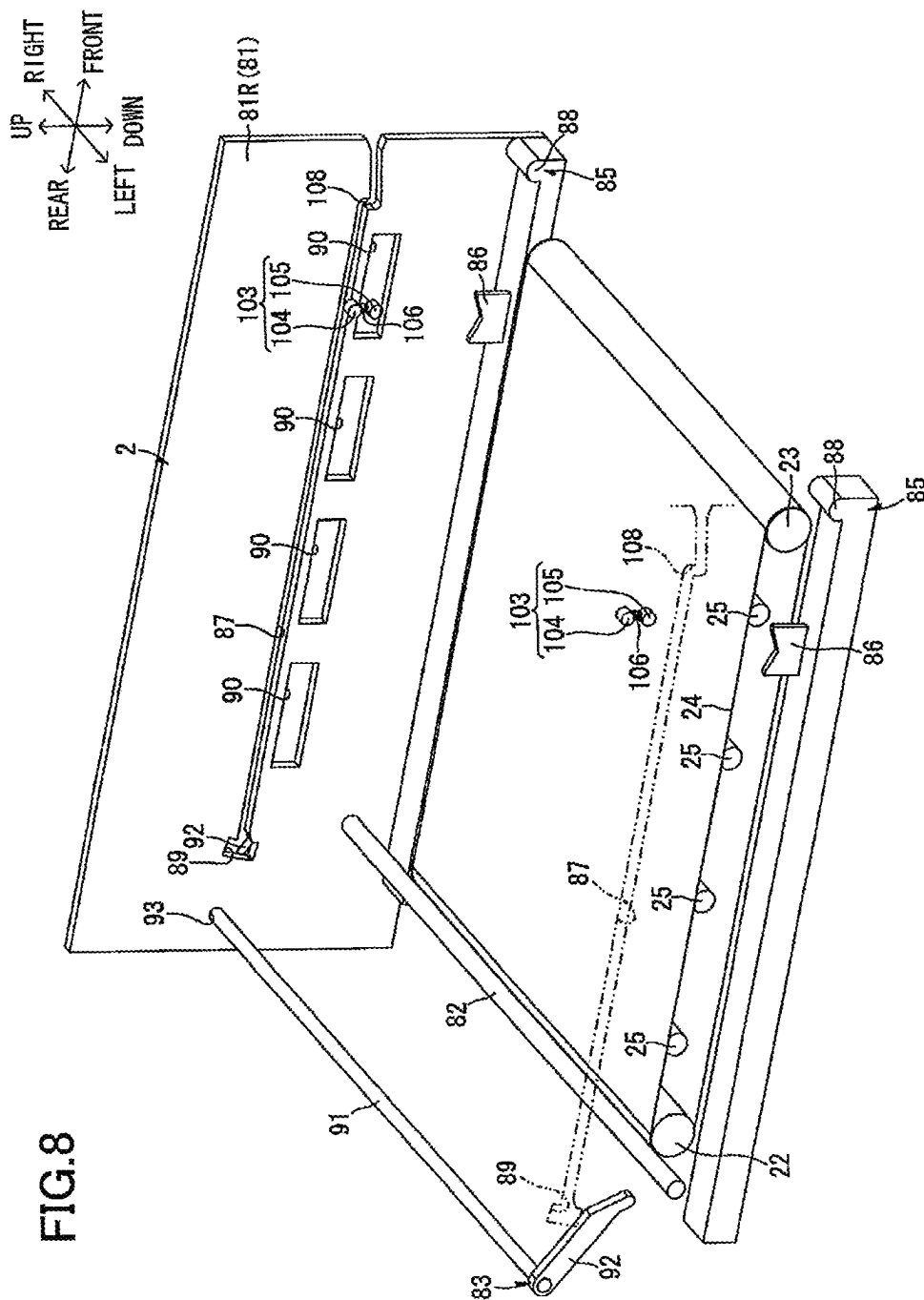
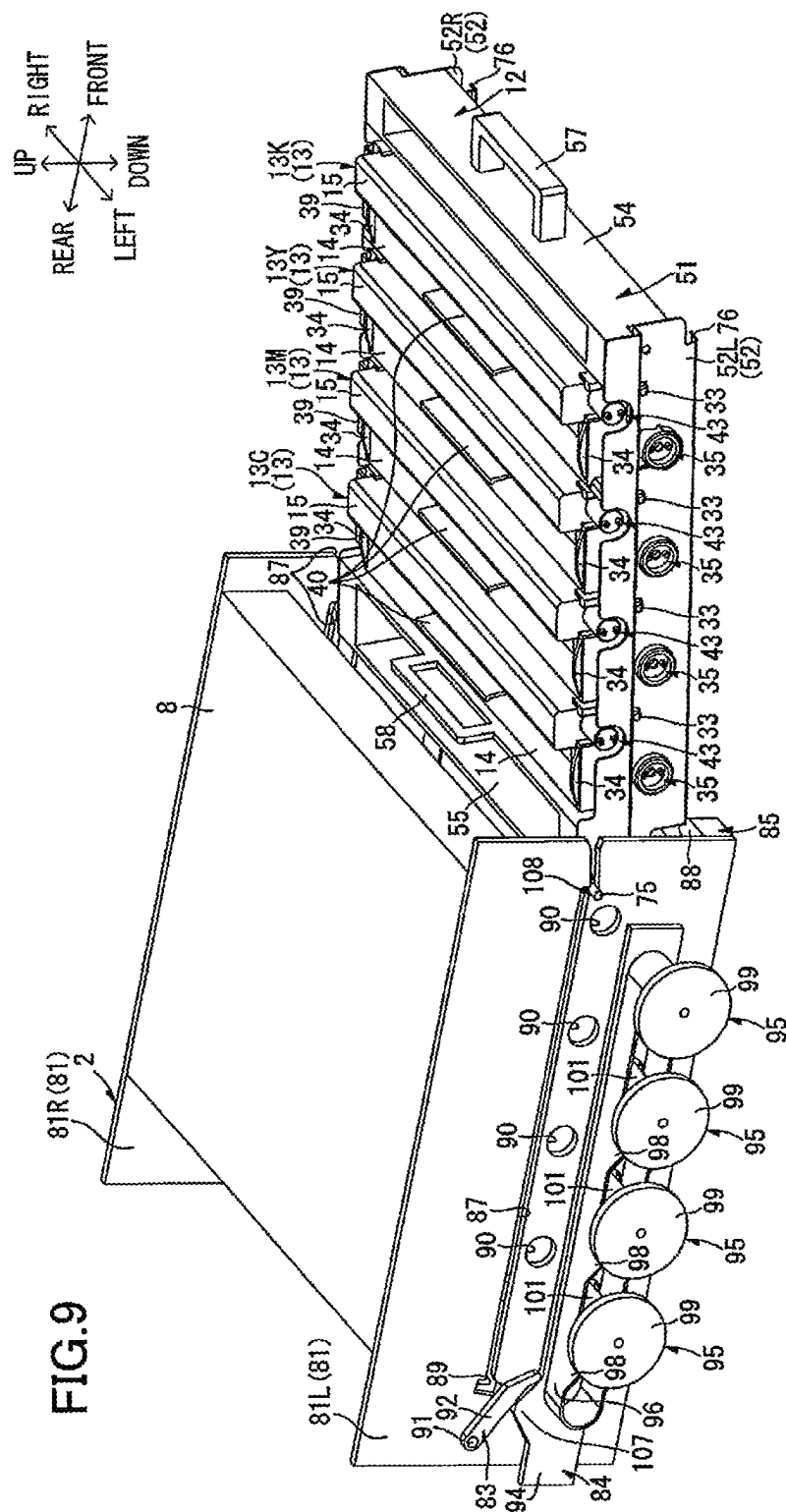


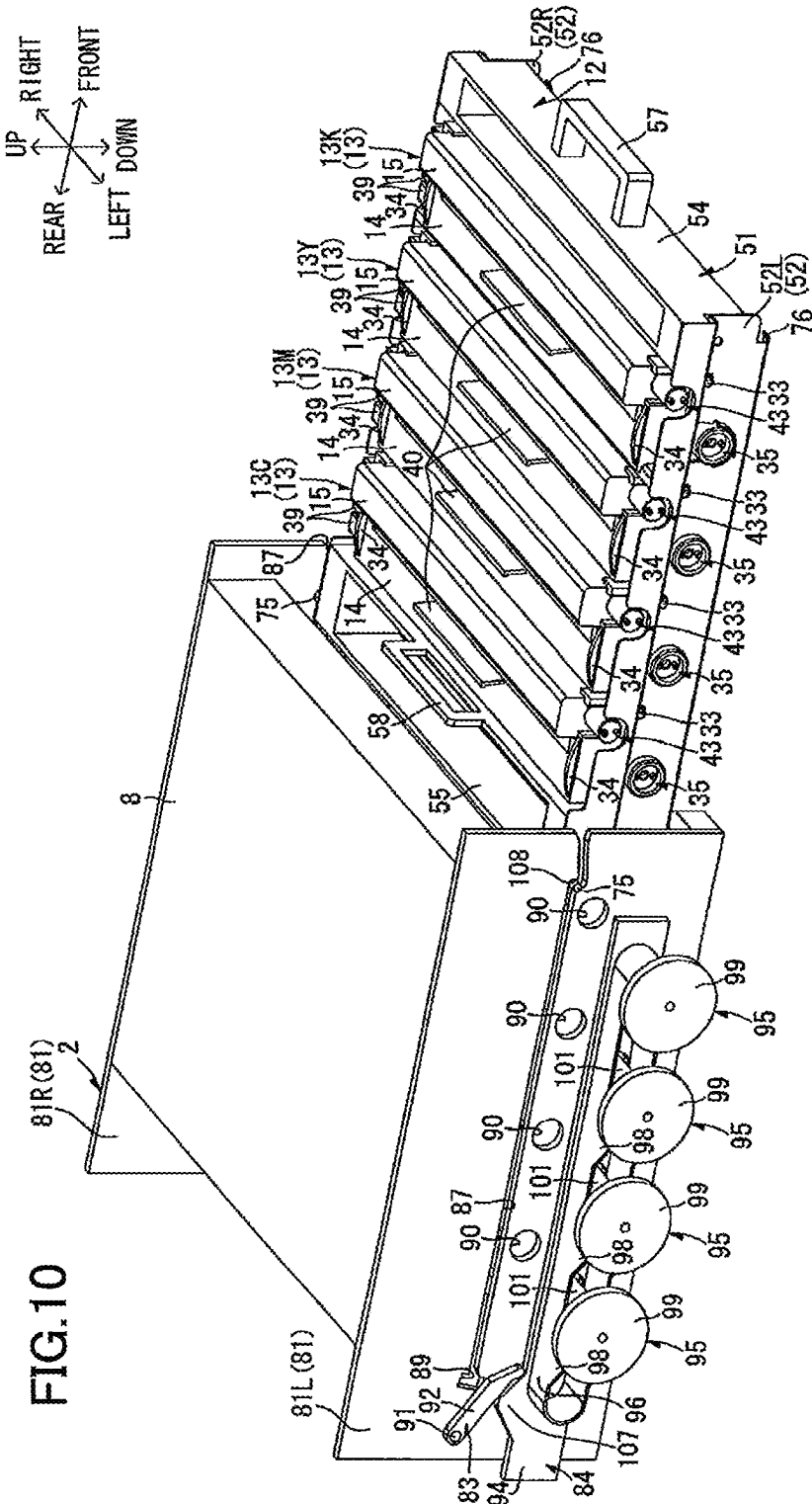
FIG. 5











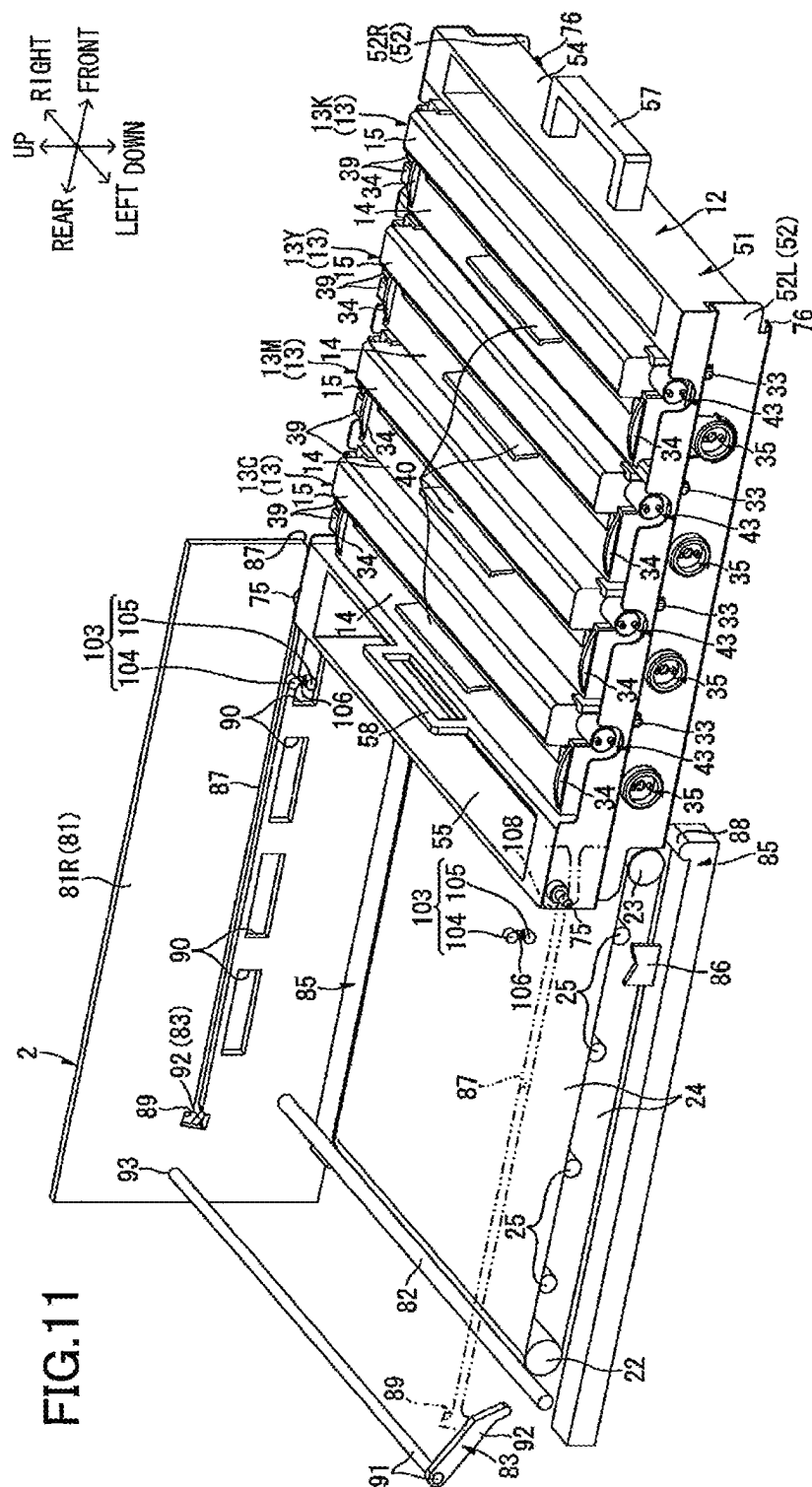


FIG. 12

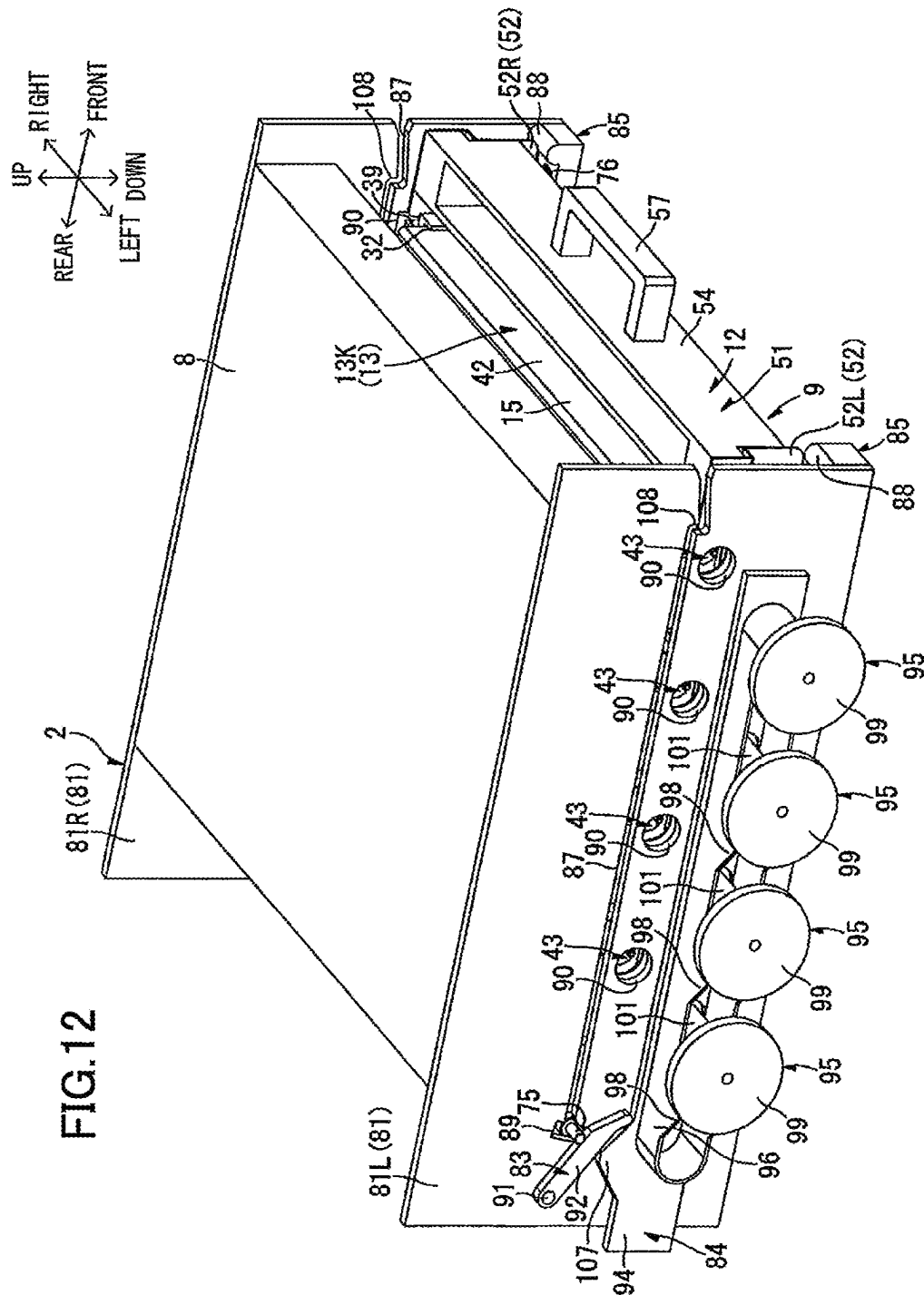
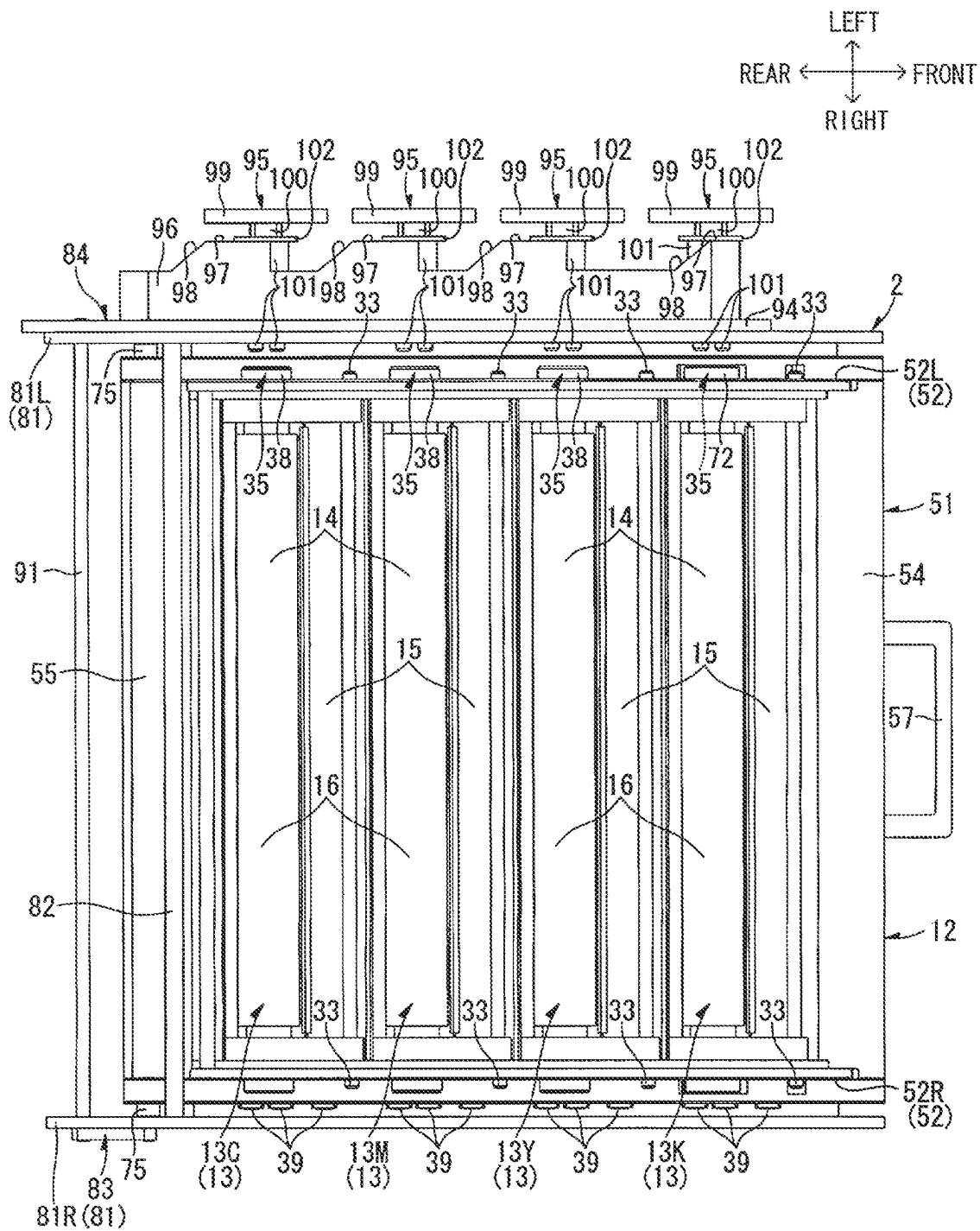
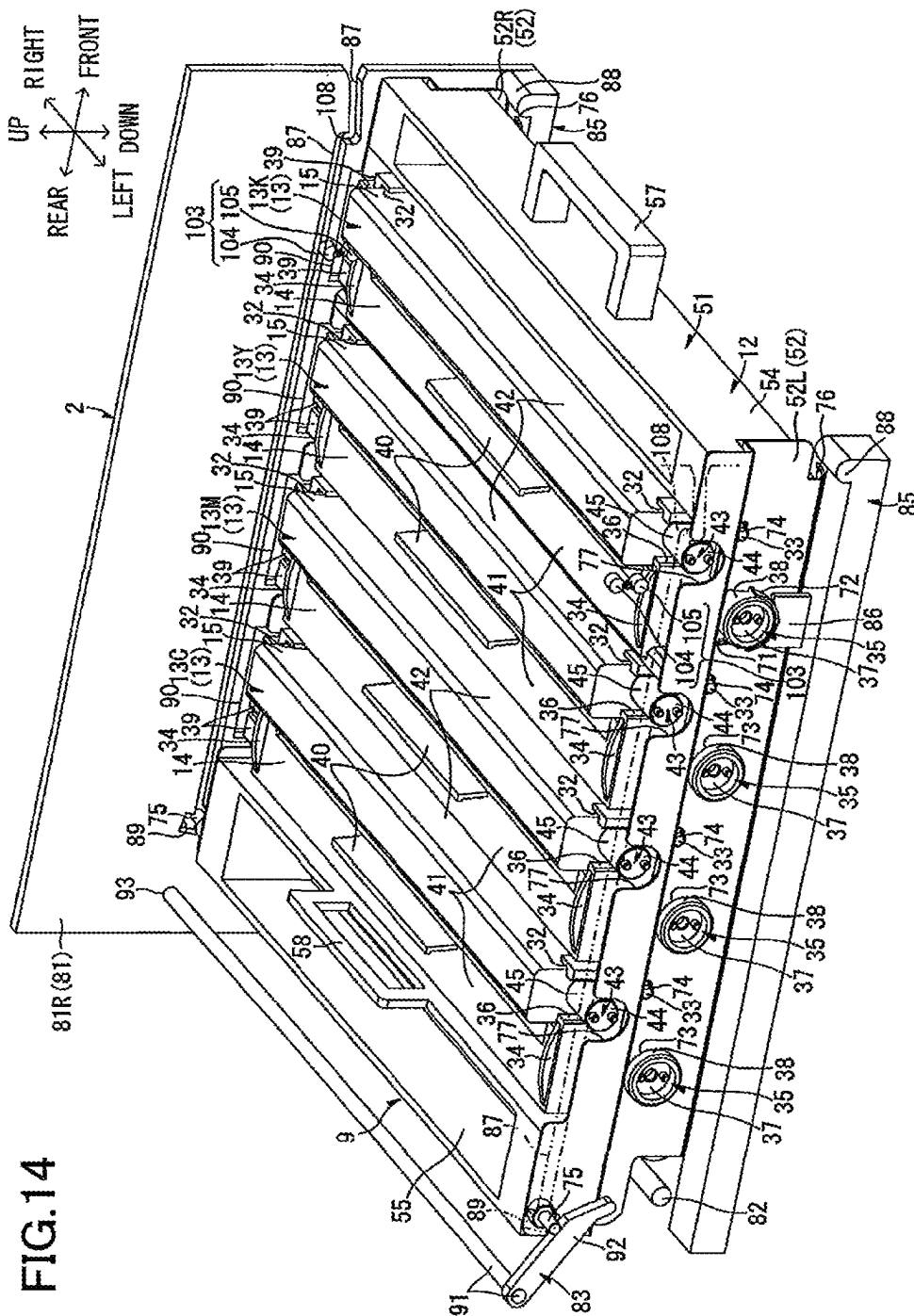


FIG. 13





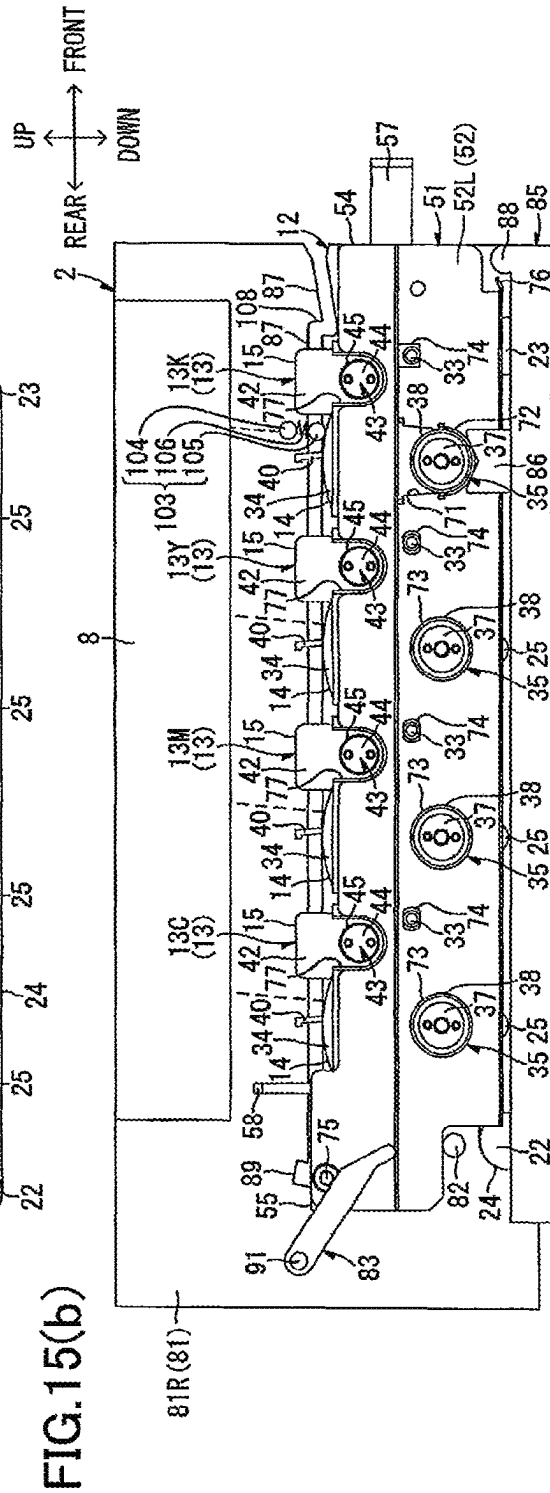
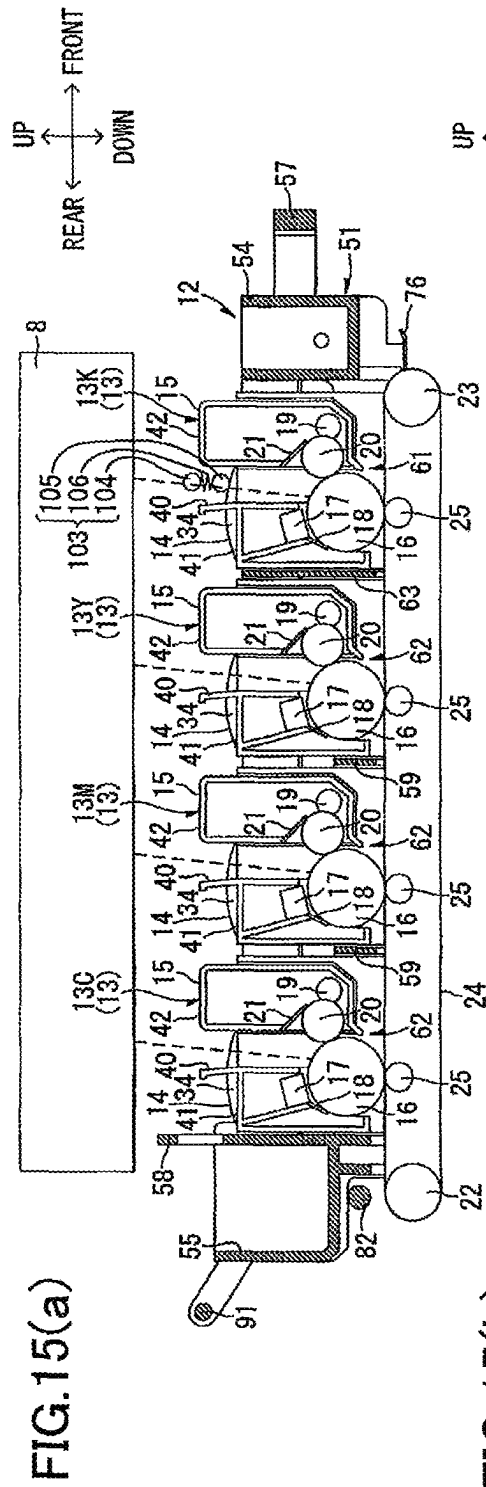


FIG. 16

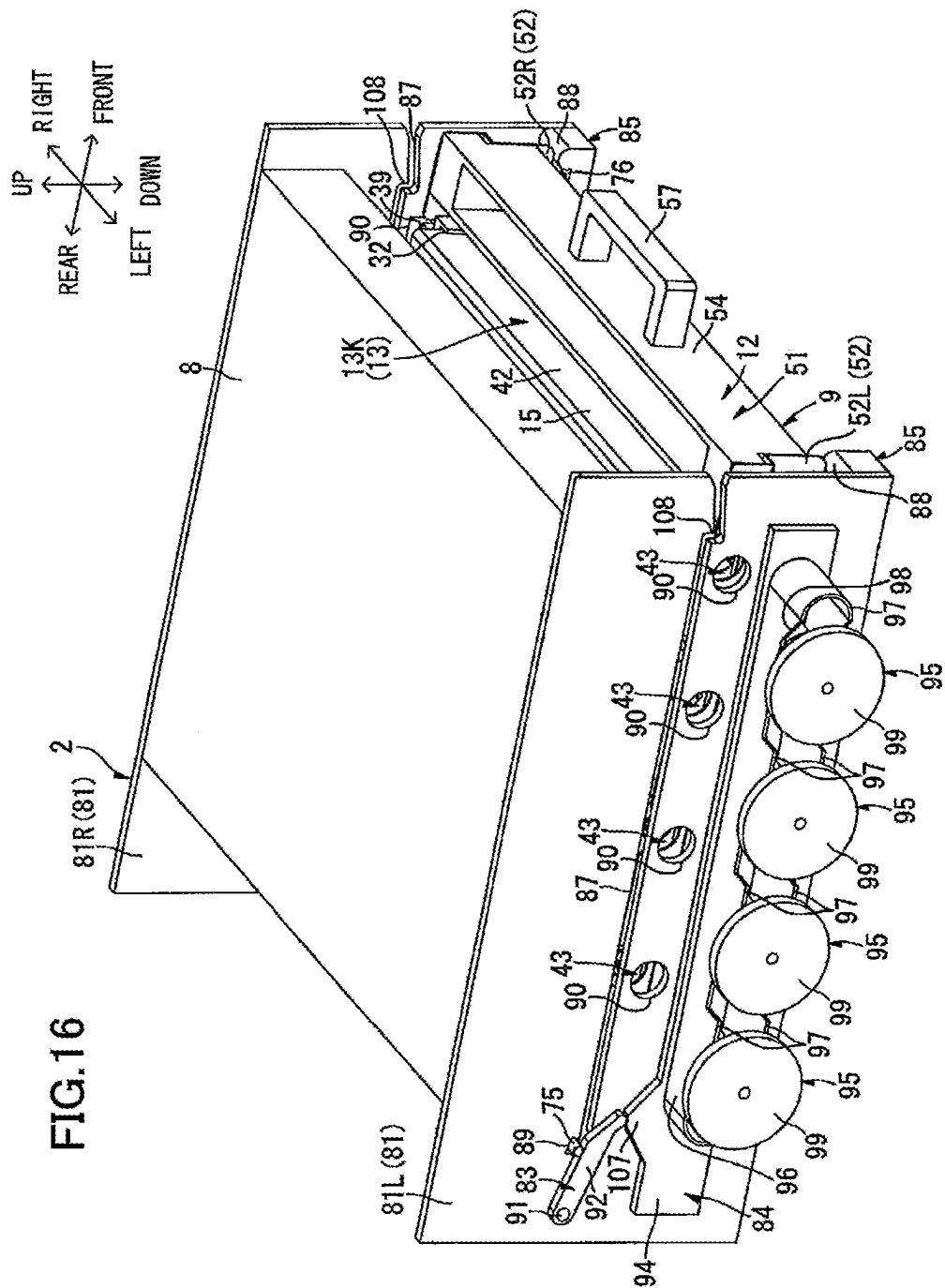
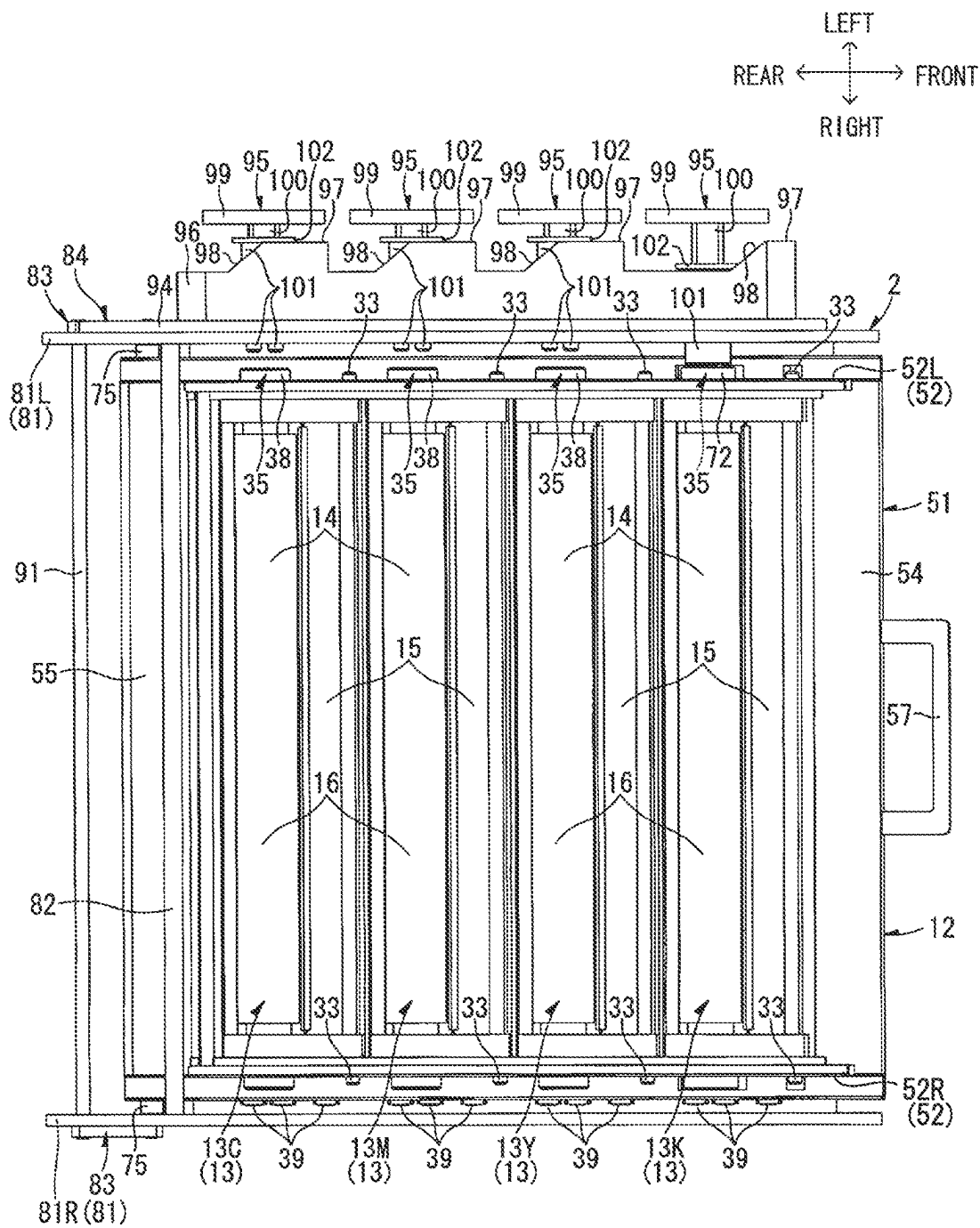


FIG.17



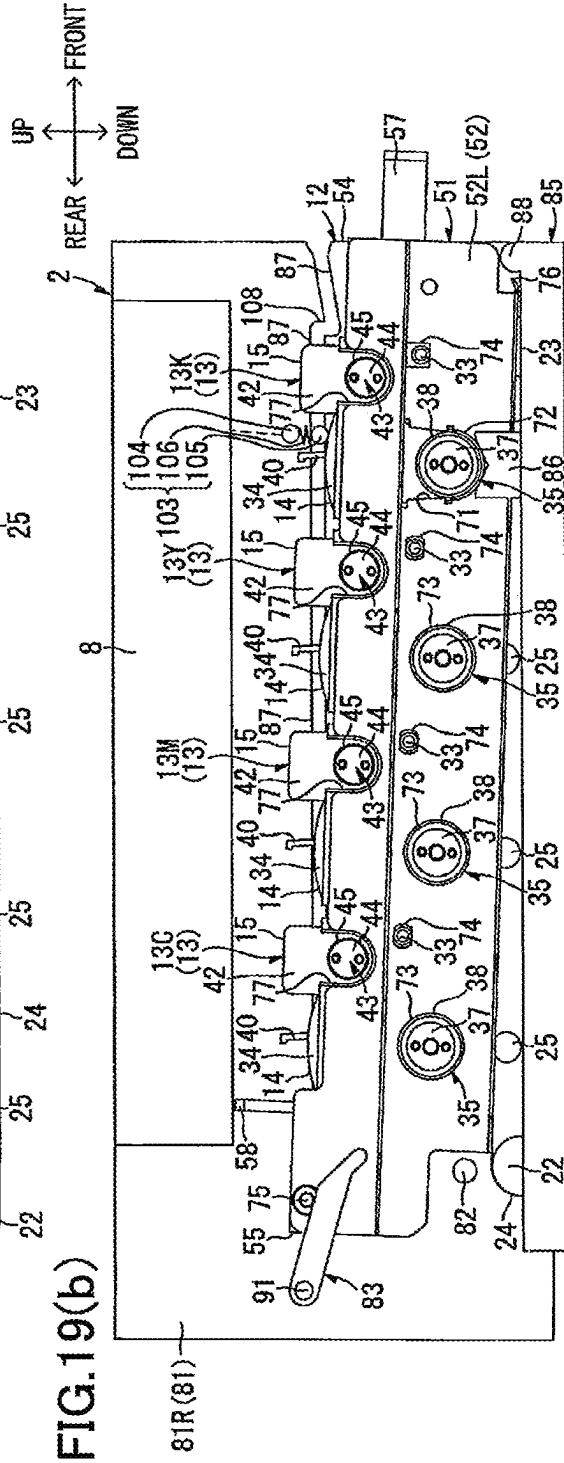
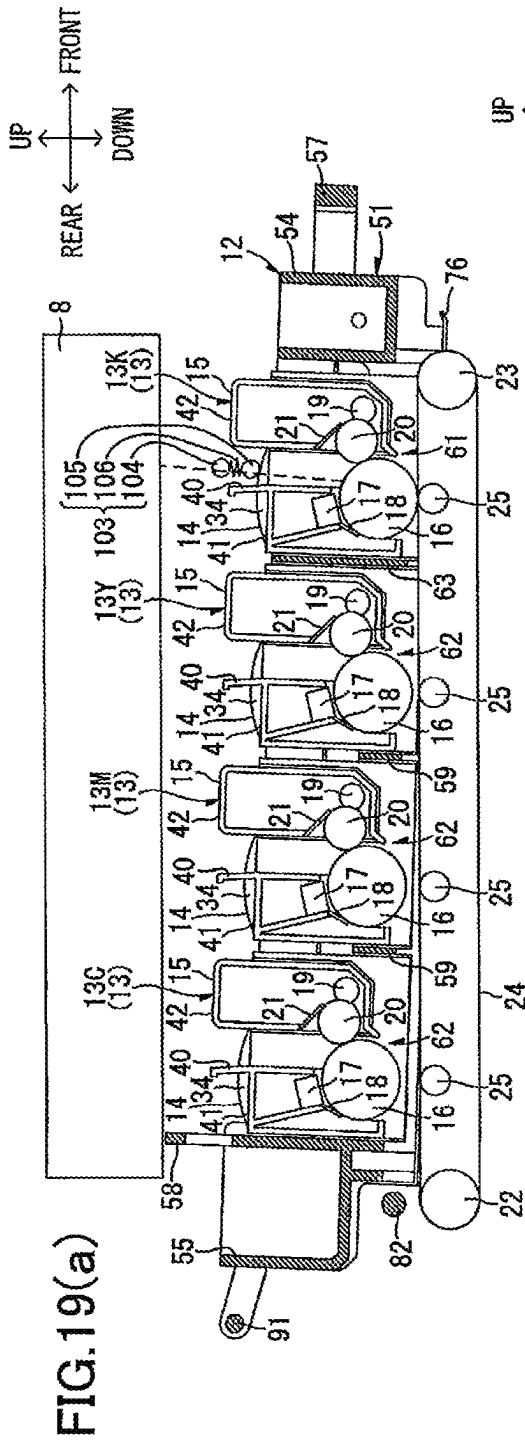


FIG. 20

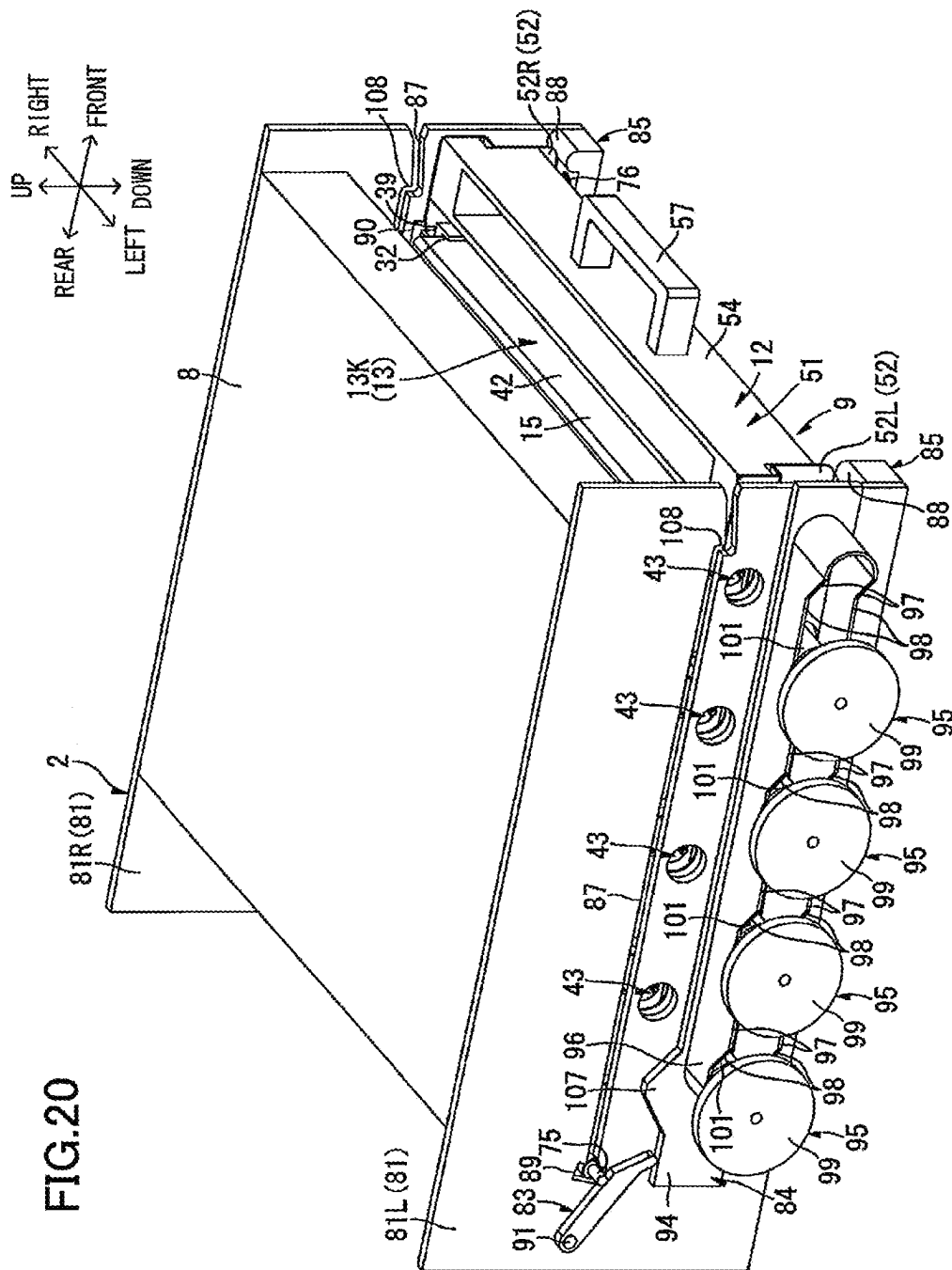


FIG.21

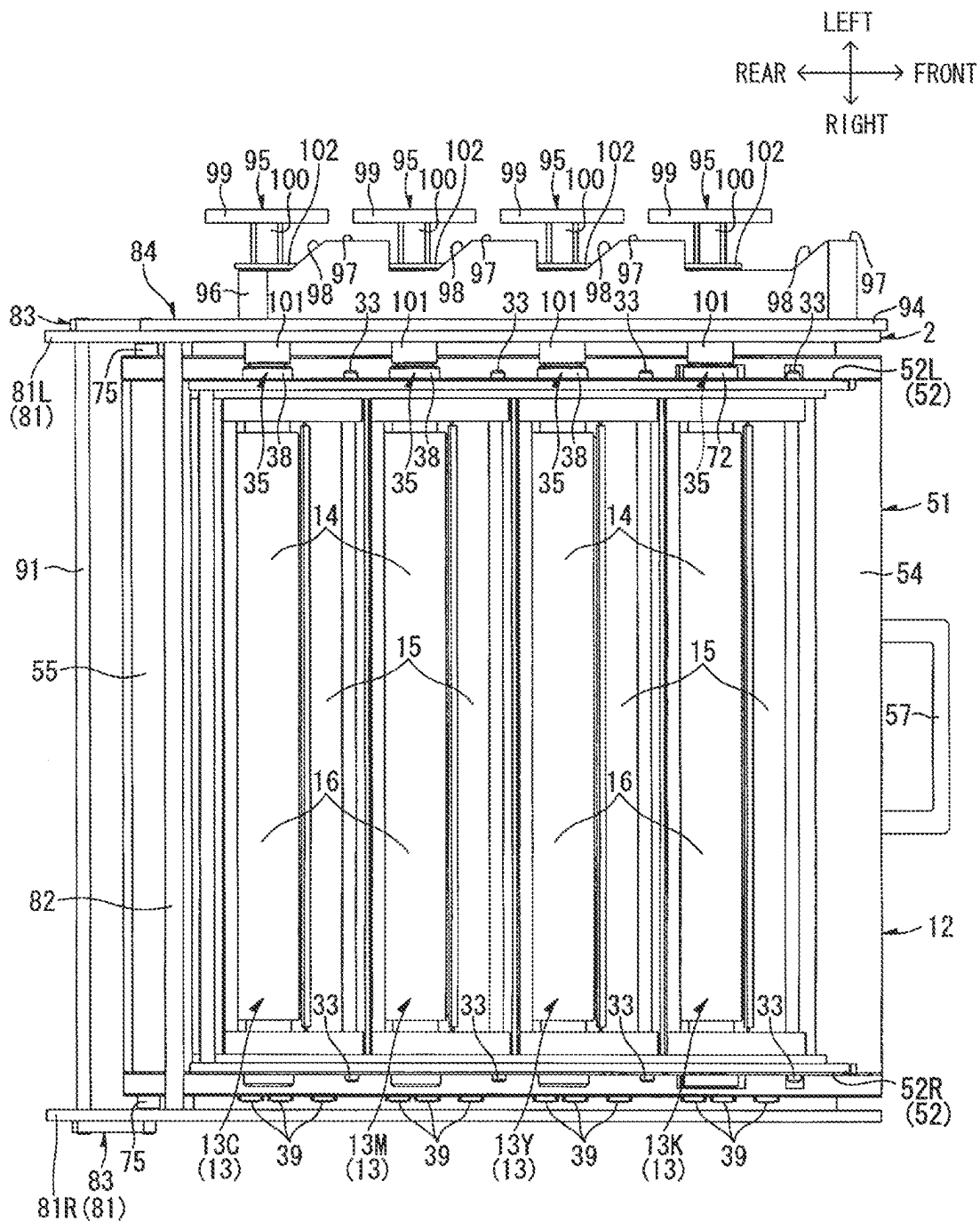
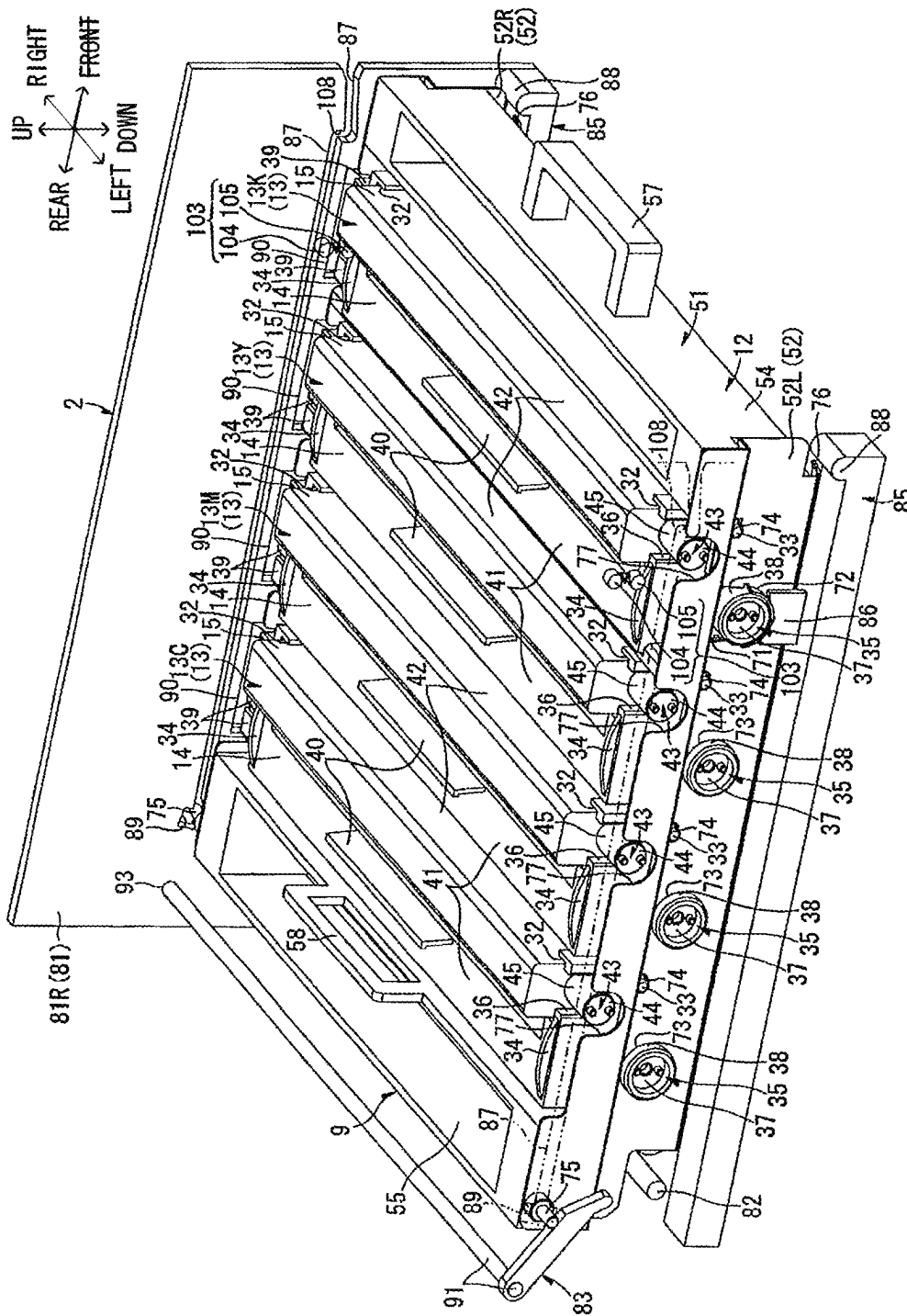


FIG.22



1

IMAGE FORMING APPARATUS INCLUDING COUPLING MEMBER SELECTIVELY COUPLED TO PHOTSENSITIVE DRUM

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/752,219 filed May 24, 2022, which is a continuation of U.S. patent application Ser. No. 17/089,879 filed Nov. 5, 2020, now U.S. Pat. No. 11,353,823, which is a continuation of U.S. patent application Ser. No. 16/573,094 filed Sep. 17, 2019, now U.S. Pat. No. 10,831,152, which is a continuation of U.S. patent application Ser. No. 16/130,438 filed Sep. 13, 2018, now U.S. Pat. No. 10,444,700, which is a continuation of U.S. patent application Ser. No. 15/275,905 filed Sep. 26, 2016, now U.S. Pat. No. 10,082,766, which is a continuation of U.S. patent application Ser. No. 14/948,591 filed Nov. 23, 2015, now U.S. Pat. No. 9,477,202, which is a continuation of U.S. patent application Ser. No. 14/053,143 filed Oct. 14, 2013, now U.S. Pat. No. 9,201,370, which is a divisional of U.S. patent application Ser. No. 12/606,660, filed Oct. 27, 2009, now U.S. Pat. No. 8,577,255, which claims priority from Japanese Patent Application No. 2008-281992 filed Oct. 31, 2008, Japanese Patent Application No. 2008-281993 filed Oct. 31, 2008, and Japanese Patent Application No. 2008-281991 filed Oct. 31, 2008. The entire contents of the above noted applications are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to an image forming apparatus.

BACKGROUND

One image forming apparatus known in the art is an electrophotographic color printer having a conveying belt disposed in confrontation with four photosensitive members corresponding to the four colors black, yellow, magenta, and cyan. However, the photosensitive members in this image-forming device are constantly in contact with the conveying belt. Therefore, even during operations to form images using only black, for example, the photosensitive members corresponding to the remaining colors yellow, magenta, and cyan remain in contact with the conveying belt and, consequently, wear more quickly.

Therefore, color image recorders have been proposed as image-forming devices capable of preventing such unnecessary wear and deterioration of the photosensitive members. For example, a color image recorder configures yellow, magenta, and cyan image-forming parts as a single color image-forming unit, and a black image-forming part as a single image-forming unit. With this image recorder, the color image-forming unit is separated from the conveying belt when forming black images.

However, in the color image recorder described above, the black image-forming part and color image-forming unit are operated independently. This configuration can potentially lead to fluctuations in the gap between the black image-forming part and the color image-forming unit during operations, making it difficult to maintain a uniform interval between neighboring photosensitive members. Maintaining this uniform interval between neighboring photosensitive

2

members is extremely important for preventing color registration problems in color printers.

SUMMARY

In view of the foregoing, it is an object of the present invention to provide an image-forming apparatus capable of separating photosensitive members from an endless belt while preserving the interval between adjacent photosensitive members.

In order to attain the above and other objects, the invention provides an image forming apparatus including a main body, an endless belt, a plurality of photosensitive drums, a mode setting unit, a contacting/separating mechanism, a transmitting mechanism, and a plurality of coupling members. The plurality of photosensitive drums is disposed in confrontation with the endless belt and is arranged in a first direction. Each photosensitive drum has an axis extending in a second direction perpendicular to the first direction. The mode setting unit sets the image forming apparatus to one of a monochrome printing mode and a color printing mode. The contacting/separating mechanism controls at least one of the plurality of photosensitive drums to contact or separate from the endless in accordance with the set mode. The transmitting mechanism selectively transmits drive power to the plurality of photosensitive drums. The plurality of coupling members correspond respectively to the plurality of photosensitive drums. Each coupling member is selectively coupled to the corresponding photosensitive drum and has a coupling axis extending in the second direction of the corresponding photosensitive drum. Each coupling member is uncoupled from the corresponding photosensitive drum when the contacting/separating mechanism controls the corresponding photosensitive drum to separate from the endless belt.

BRIEF DESCRIPTION OF THE DRAWINGS

The particular features and advantages of the invention as well as other objects will become apparent from the following description taken in connection with the accompanying drawings, in which:

FIG. 1 is a side cross-sectional view of an image forming apparatus according to a preferred embodiment of the present invention;

FIG. 2 is a perspective view of a drum unit provided in the image forming apparatus;

FIG. 3 is an exploded perspective view of the drum unit shown in FIG. 2;

FIG. 4 is a perspective view of the drum unit when color photosensitive drums are mounted in a process frame;

FIG. 5 is a perspective view of the drum unit when a black photosensitive drum is mounted in the process frame shown in FIG. 4;

FIG. 6 is an explanatory diagram showing how to position the black photosensitive drum;

FIG. 7 is a perspective view illustrating an operation for mounting the drum unit in a main body of the image forming apparatus;

FIG. 8 is an explanatory diagram illustrating an interior of the main body shown in FIG. 7;

FIG. 9 is a perspective view illustrating an operation for mounting the drum unit in the main body;

FIG. 10 is an explanatory diagram illustrating an operation for mounting the drum unit in the main body;

FIG. 11 is an explanatory diagram illustrating the interior of the main body in a state shown in FIG. 10;

3

FIG. 12 is a perspective view illustrating the operation for mounting the drum unit from the state shown in FIG. 10, where the drum unit has been completely mounted in the main body;

FIG. 13 is a base view illustrating the image forming apparatus in the state shown in FIG. 12;

FIG. 14 is an explanatory diagram illustrating the interior of the main body shown in FIG. 12;

FIG. 15(a) is a perspective view illustrating the drum unit and the main body in the state shown in FIG. 12;

FIG. 15(b) is a side view illustrating the drum unit and the main body in the state shown in FIG. 12;

FIG. 16 is a perspective view illustrating the image forming apparatus when the image forming apparatus is set to a monochrome mode;

FIG. 17 is a base view illustrating the image forming apparatus when the image forming apparatus is set to the monochrome mode;

FIG. 18 is an explanatory diagram illustrating the interior of the main body when the image forming apparatus is set to the monochrome mode;

FIG. 19(a) is a perspective view illustrating the drum unit and the main body when the image forming apparatus is set to the monochrome mode;

FIG. 19(b) is a side view illustrating the drum unit and the main body when the image forming apparatus is set to the monochrome mode;

FIG. 20 is a perspective view illustrating the image forming apparatus when the image forming apparatus is set to a color mode;

FIG. 21 is a base view illustrating the image forming apparatus when the image forming apparatus is set to the color mode; and

FIG. 22 is an explanatory diagram illustrating the interior of the main body when the image forming apparatus is set to the color mode.

DETAILED DESCRIPTION

As shown in FIG. 1, a printer 1 serving as a preferred embodiment of the image-forming apparatus according to the present invention is a horizontal tandem-type color laser printer that includes a main casing 2 as the main body of the printer, and, within the main casing 2, a feeding unit 3 for supplying sheets of a paper P to be printed, and an image-forming unit 4 for forming images on the sheets of paper P supplied from the feeding unit 3. The terms “up”, “down”, “upper”, “lower”, “above”, “below”, “beneath”, “right”, “left”, “front”, “rear” and the like will be used throughout the description assuming that the printer 1 is disposed in an orientation in which it is intended to be used. In use, the printer 1 is disposed as shown in FIG. 1.

The main casing 2 has a box shape that is substantially rectangular in a side view. The image-forming unit 4 is accommodated in the main casing 2. A front cover 5 is provided on one side wall of the main casing 2 for exposing the inside of the main casing 2 in order to mount or remove a process frame 12 described later.

In the following description, the side of the color laser printer 1 on which the front cover 5 is provided (right side in FIG. 1) will be referred to as the front side, and the opposite side (left side in FIG. 1) as the rear side. The left and right sides of the color laser printer 1 will be based on a user's perspective when viewing the color laser printer 1 from the front. Hence, the near side of the color laser printer 1 in FIG. 1 is the left side, and the far side is the right side.

4

The feeding unit 3 includes a paper tray 6 for accommodating sheets of the paper P. The paper tray 6 is detachably mounted in the bottom section of the main casing 2. A feeding roller 7 is disposed above the front end of the paper tray 6 for feeding sheets of paper P from the paper tray 6 to the image-forming unit 4 along a U-shaped feeding path (not shown).

The feeding roller 7 rotates to feed sheets of paper P accommodated in the paper tray 6 onto the feeding path one sheet at a time. The sheets of paper P are subsequently conveyed from the feeding path to the image-forming unit 4 so as to pass between four photosensitive drums 16 and a conveying belt 24 described later.

The image-forming unit 4 includes a scanning unit 8, a process unit 9, a transfer unit 10, and a fixing unit 11. The scanning unit 8 is disposed above the main casing 2. The scanning unit 8 irradiates laser beams toward the four photosensitive drums 16 based on image data to expose the surfaces of the corresponding photosensitive drums 16.

The process unit 9 is disposed below the scanning unit 8 and above the feeding unit 3. The process unit 9 is detachably mounted on the main casing 2. The process unit 9 includes the single process frame 12, and four process cartridges 13 corresponding to the four printing colors.

The process frame 12 is slidably supported in the main casing 2 and can be slid into or out of the main casing 2 in the front-to-rear direction. The process cartridges 13 are mounted in the process frame 12 in a juxtaposed arrangement in the front-to-rear direction (mounting direction). More specifically, the process cartridges 13 support a black process cartridge 13K, a yellow process cartridge 13Y, a magenta process cartridge 13M, and a cyan process cartridge 13C arranged from the front side to the rear side in the sequence given. Each process cartridge 13 is provided with a drum unit 14, and a developer cartridge 15.

Each drum unit 14 includes a photosensitive drum 16, a Scorotron charger 17, and a cleaning blade 18. The photosensitive drum 16 is oriented with its axis along the left-to-right direction. Specifically, the photosensitive drum 16 of the black process cartridges 13K has an axis extending in a first axial direction parallel to the left-to-right direction. Each of photosensitive drums 16 of the yellow process cartridge 13Y, the magenta process cartridge 13M, and the cyan process cartridge 13C has an axis extending in a second axial direction parallel to the first axial direction. The black photosensitive drum 16 is disposed at a position nearest to one end of the process unit 9 that is upstream in the mounting direction.

The Scorotron charger 17 is disposed diagonally above and rearward of the photosensitive drum 16 and confronts but does not contact the photosensitive drum 16. The cleaning blade 18 is disposed to the rear of the photosensitive drum 16 and confronts and contacts the photosensitive drum 16.

The developer cartridge 15 is disposed on the front side of the corresponding drum unit 14 and includes a supply roller 19, a developing roller 20, and a thickness-regulating blade 21.

The developing roller 20 is disposed so as to contact the front side of the photosensitive drum 16. The supply roller 19 is disposed on the front side of the developing roller 20. The thickness-regulating blade 21 is disposed above the developing roller 20. The space formed in the developer cartridge 15 above these components serves to accommodate toner in the corresponding color.

With the process cartridge 13, toner accommodated in the developer cartridge 15 is supplied onto the supply roller 19,

5

which in turn supplies toner to the developing roller 20. At the same time, the toner is positively tribocharged between the supply roller 19 and developing roller 20.

As the developing roller 20 rotates, the thickness-regulating blade 21 regulates the toner carried on the surface of the developing roller 20 to a prescribed thickness so that the developing roller 20 carries a uniform thin layer of toner.

In the meantime, the Scorotron charger 17 applies a uniform charge of positive polarity to the surface of the photosensitive drum 16 while the photosensitive drum 16 rotates. Subsequently, the scanning unit 8 irradiates a laser beam (indicated by dash-line in FIG. 1) through the gap formed between the photosensitive drum unit 14 and developer cartridge 15 to expose the surface of the photosensitive drum 16 in a high-speed scan. In this way, the scanning unit 8 forms an electrostatic latent image on the surface of the photosensitive drum 16 corresponding to an image to be formed on the paper P.

As the photosensitive drum 16 continues to rotate, the positively charged toner carried on the surface of the developing roller 20 is supplied to the electrostatic latent image formed on the surface of the photosensitive drum 16, thereby developing the electrostatic latent image into a visible toner image through reverse development.

The transfer unit 10 is disposed in the main casing 2 above the feeding unit 3 and below the process unit 9 and extends in the front-to-rear direction. The transfer unit 10 includes a drive roller 22, a follow roller 23, the conveying belt 24, and four transfer rollers 25.

The drive roller 22 and follow roller 23 are parallel to each other and separated in the front-to-rear direction. The endless conveying belt 24 and is mounted around the drive roller 22 and follow roller 23. The transfer rollers 25 are disposed inside the conveying belt 24 at positions opposing the photosensitive drums 16 with the conveying belt 24 interposed therebetween. Positions between each photosensitive drum 16 and the corresponding transfer roller 25 are referred to as transfer positions.

The upper portion of the conveying belt 24 moves rearward for conveying a sheet of paper P supplied from the feeding unit 3 sequentially through each transfer position between the photosensitive drums 16 and transfer rollers 25. As the sheet is conveyed on the conveying belt 24, toner images in each color carried on the respective photosensitive drums 16 are sequentially transferred onto the sheet to form a color image.

The fixing unit 11 is disposed to the rear of the transfer unit 10 and includes a heating roller 26, and a pressure roller 27 in confrontation with the heating roller 26. After a color image has been transferred onto the sheet of paper P in the transfer unit 10, the image is fixed to the sheet by a combination of heat and pressure as the sheet passes between the heating roller 26 and pressure roller 27 in the fixing unit 11.

After the toner image has been fixed to the paper P, the sheet is conveyed along a U-shaped discharge path (not shown) toward a pair of discharge rollers 28 disposed at the downstream end of the path. The discharge rollers 28 discharge the sheet onto a discharge tray 29 formed on the top surface of the main casing 2.

As shown in FIGS. 2 and 3, each process cartridge 13 is provided with a pair of left and right side walls 31 disposed substantially parallel to each other but separated in the left-to-right direction, the photosensitive drum unit 14 spanning between the side walls 31, and the developer cartridge 15 detachably supported in the side walls 31.

6

Each of the side walls 31 is a flat plate that is rectangular in a side view and has a thickness in the left-to-right direction. Hereinafter, when distinguishing between the side walls 31 disposed on the left and right sides, the side wall 31 on the left side will be referred to as the left side wall 31L, while the side wall 31 on the right side will be referred to as the right side wall 31R. Developer guide grooves 32 are formed in the inner surfaces of both side walls 31 for receiving both left and right ends of the developer cartridges 15. Further, stoppers 33 are provided on the outer surfaces of both side walls 31, while protrusions 34 are formed on the top edges of the side walls 31.

The developer guide grooves 32 are formed as cutouts in the front portions of both side walls 31 and are substantially rectangular in shape in a side view, extending from the top edges of the side walls 31 to a point near the bottom edges.

The stoppers 33 are cylindrical ribs that protrude outward from the outer surfaces of both side walls 31 in the respective left and right directions. The stoppers 33 are positioned near the front edges of the side walls 31 and in substantially the vertical center thereof.

The protrusions 34 are plates extending upward from the top edges of the side walls 31. Each protrusion 34 is arc-shaped in a side view, sharing a central axis with the respective photosensitive drum 16. Laterally, the protrusions 34 extend from the rear edges of the side walls 31 to the centers thereof.

In addition, a drum coupling 35 is provided near the lower rear corner of the left side wall 31L for inputting a drive force from the main casing 2 side. A developer coupling support groove 36 is formed in the top edge of the left side wall 31L near the front side thereof for receiving a developer coupling (described later).

The drum coupling 35 includes a drum coupling cover 38, and a female drum coupling member 37 rotatably supported by the drum coupling cover 38. The drum coupling cover 38 is cylindrically shaped and protrudes leftward from the left surface of the left side wall 31L. The female drum coupling member 37 is coupled to the left end of the photosensitive drum 16 so as to be unable to rotate relative to the photosensitive drum 16.

The developer coupling support groove 36 is a cutout formed in the top edge of the left side wall 31L at a position overlapping the developer guide grooves 32 when projected from left to right. The developer coupling support groove 36 has a U-shape in a side view with an open top. A drum support member (not shown) is also provided on the right side wall 31R for supporting the right end of the photosensitive drum 16 so as to be incapable of rotating relative to the photosensitive drum 16. The drum support member has the same cylindrical shape as the drum coupling cover 38 and protrudes rightward from the right surface of the right side wall 31R.

A plurality of electrodes 39 is provided on the right side wall 31R. Body-side electrodes (not shown) provided in the main casing 2 are connected to the electrodes 39 for supplying power to the process cartridge 13.

Each of the drum units 14 includes the photosensitive drum 16, Scorotron charger 17, cleaning blade 18, and a drum partition 41 accommodating these components. The photosensitive drum 16 is rotatably supported between the corresponding side walls 31. The Scorotron charger 17 spans between the side walls 31 at a position diagonally above and rearward of the corresponding photosensitive drum 16 (see FIG. 1). The cleaning blade 18 also spans between the corresponding side walls 31 on the rear side of the photosensitive drum 16 (see FIG. 1).

Each drum partition **41** has a cylindrical shape that is open on the bottom. The drum partition **41** spans between the corresponding side walls **31** and is formed integrally therewith. The side walls **31** and the drum partition **41** for covering the photosensitive drum **16**. A grip part **40** is integrally formed on the top surface of each drum unit **14**.

The developer cartridge **15** includes the supply roller **19**, the developing roller **20**, and a developer frame **42** for accommodating these components. The supply roller **19** and developing roller **20** are disposed adjacent to each other in the bottom end of the developer frame **42**, with the supply roller **19** on the front side and the developing roller **20** on the rear side, and are rotatably supported in the developer frame **42** (see FIG. 1).

The developer frame **42** extends in the left-to-right direction and is box-shaped, with an opening formed in the lower rear side at a position confronting the developing roller **20**.

A developer coupling **43** is provided on the left end of the developer frame **42** in the vertical center thereof for inputting a drive force from the main casing **2** side to the supply roller **19** and developing roller **20**.

The developer coupling **43** includes a developer coupling cover **45**, and a female developer coupling member **44** rotatably supported by the developer coupling cover **45**. The developer coupling cover **45** has a cylindrical shape and protrudes leftward from the left endface of the developer frame **42**. The female developer coupling member **44** is coupled with a gear part (not shown) for driving the supply roller **19** and developing roller **20**.

In order to mount the developer cartridge **15** in the photosensitive drum unit **14**, the operator grips the top of the developer frame **42** so that the developing roller **20** is positioned on the bottom of the developer cartridge **15** and the developer coupling **43** is positioned on the left. The operator aligns the developer cartridge **15** above the photosensitive drum unit **14** for fitting the left and right ends of the developer cartridge **15** into the developer guide grooves **32** of the side walls **31** and for fitting the developer coupling **43** into the developer coupling support groove **36**, and inserts the developer cartridge **15** downward into the photosensitive drum unit **14**.

At this time, the left and right ends of the developer cartridge **15** are fitted into the developer guide grooves **32** and guided downward therein. The lower edges of the left and right ends of the developer cartridge **15** contact the lower edges of the developer guide grooves **32** while the developer coupling **43** is fitted into the developer coupling support groove **36**. This completes mounting of the developer cartridge **15** in the photosensitive drum unit **14**.

Next, details of the process frame **12** will be described. The process frame **12** includes a cartridge support frame **51** for supporting all four of the process cartridges **13** from the bottom thereof, and a pair of side plates **52** provided on the left and right sides of the cartridge support frame **51**.

The cartridge support frame **51** is a framework formed to encompass the peripheries of the four process cartridges **13**. The cartridge support frame **51** is integrally provided with a pair of left and right frame side walls **53**, a front beam **54**, and a rear beam **55**. The cartridge support frame **51** also has a black partitioning plate **61** spanning between the frame side walls **53** at a position between the front beam **54** and rear beam **55**.

The black partitioning plate **61** serves to partition the space between the front beam **54** and the rear beam **55** in the cartridge support frame **51** into a black mounting portion **62** for the black process cartridge **13K**, and a color mounting portion **63** to the rear of the black mounting portion **62** for

the three remaining process cartridges **13**. In other words, the black mounting portion **62** is defined by the frame side walls **53**, the front beam **54**, and the black partitioning plate **61**, while the color mounting portion **63** is defined by the frame side walls **53**, the rear beam **55**, and the black partitioning plate **61**.

As shown in FIG. 3, two partitioning plates **59** span in the left-to-right direction between the frame side walls **53** at positions over the color mounting portion **63**. Each partitioning plates **59** also extends vertically with the left and right ends thereof connected to the corresponding frame side walls **53** at positions between neighboring drum support grooves **56** (described later). Hence, the space encompassed by the front beam **54**, rear beam **55**, and the pair of left and right frame side walls **53** is divided into four substantially equal intervals in the front-to-rear direction by the black partitioning plate **61** and the partitioning plates **59**.

The side walls **31**, developer guide grooves **32**, and the black partitioning plate **61** all extend farther upward than the partitioning plates **59**. The frame side walls **53** are arranged parallel to each other and are separated in the left-to-right direction. Hereinafter, the frame side wall **53** on the left side will be referred to as the left frame side wall **53L**, and the frame side wall **53** on the right side the right frame side wall **53R** when it is necessary to distinguish between the two.

Four drum support grooves **56** are formed at substantially regular intervals in the front-to-rear direction in each of the frame side walls **53**. The drum support grooves **56** are substantially U-shaped in a side view so as to be open on the top and are formed at positions corresponding to the drum couplings **35** of the process cartridges **13** or drum support members (not shown). The drum support grooves **56** formed in the left frame side wall **53L** are formed in a shape for receiving the drum coupling covers **38**, while the drum support grooves **56** in the right frame side wall **53R** are formed in a shape for receiving the drum support members.

A stopper support groove **64** is formed in each of the frame side walls **53** in the portion of the frame side walls **53** adjacent to the black mounting portion **62**. The stopper support grooves **64** are formed in front of the corresponding drum support grooves **56** at positions aligned with the stoppers **33** of the black process cartridge **13K**. The stopper support grooves **64** are substantially U-shaped in a side view with an open top for receiving the stoppers **33**.

A cartridge support rail **65** is formed across the entire front-to-rear length on the lower edge of each frame side wall **53** and protrudes inward therefrom. The cartridge support rails **65** are formed at positions for receiving contact from the side walls **31** of the process cartridges **13** when the process cartridges **13** are mounted in the process frame **12**.

The front beam **54** spans between the front ends of the frame side walls **53** in the left-to-right direction and has a substantially U-shaped cross section. A first frame handle **57** is integrally formed on the front surface of the front beam **54**.

Similarly, the rear beam **55** spans between the rear ends of the frame side walls **53** in the left-to-right direction and has a substantially U-shaped cross section. A second frame handle **58** is integrally formed on the top surface of the front side of the rear beam **55**.

The side plates **52** are configured of metal plates formed through punching and pressing processes. Hereinafter, the side plate **52** on the left side will be referred to as a left side plate **52L** and the side plate **52** on the right side the right side plate **52R** when distinguishing between the two. The left side plate **52L** and right side plate **52R** are disposed parallel to each other. The side plates **52** extend in the front-to-rear

direction and are substantially rectangular in a side view. The front and rear ends of the side plates 52 confront the front beam 54 and rear beam 55, respectively, in the left-to-right direction.

The side plates 52 are bent at a vertical midpoint in substantially the shape of a crank in a front view, with the top portion positioned farther outside the bottom portion in the left-to-right direction. Specifically, when projected downward, the top portions of the side plates 52 are positioned farther outside the left and right ends of the drum couplings 35 and the drum support members (not shown) of the process cartridges 13 in the respective left and right directions.

A black support hole 71 is formed in the lower portion of each side plate 52 in the region adjacent to the black mounting portion 62 and in a position corresponding to the drum support groove 56 of the respective frame side wall 53. Each black support hole 71 is substantially U-shaped in a side view with an open top. The black support hole 71 spans from the lower part of the side plate 52 to the bent portion thereof.

A black support part 72 is formed on the bottom edge of each black support hole 71. Each black support part 72 follows an arcing shape and extends outward from the black support hole 71 in the respective left and right direction. The black support part 72 is formed integrally with the respective side plate 52 and functions to support the left and right ends of the respective drum coupling 35 and drum support member (not shown) from the bottom thereof.

Three drum retaining holes 73 that are circular in a side view are formed in the lower regions of both side plates 52 in the portion adjacent to the color mounting portion 63 and at positions corresponding to the drum support grooves 56 of the frame side walls 53.

The black support hole 71 and drum retaining holes 73 in each side plate 52 are juxtaposed at substantially regular intervals in the front-to-rear direction and are formed so as to receive the drum coupling covers 38 and drum support members (not shown).

Stopper fitting holes 74 are formed in each of the side plates 52 for receiving the stoppers 33 provided on the process cartridges 13. Four of the stopper fitting holes 74 are formed in each side plate 52 at regular intervals in the front-to-rear direction. The forwardmost stopper fitting hole 74 is a cutout that extends from the lower portion of the side plate 52 to the bent portion and is rectangular in a side view. This stopper fitting hole 74 corresponds to the stopper 33 of the black process cartridge 13K. The remaining stopper fitting holes 74 corresponding to the other process cartridges 13, i.e., the non-black (color) process cartridges 13Y, 13C, and 13M, are elongated holes extending in the front-to-rear direction in a side view.

Each side plate 52 includes a protruding part 75 on the top portion thereof, protruding outward in the left or right direction, and a rail part 76 extending along the bottom edge of the side plate 52 in the front-to-rear direction.

The protruding parts 75 constitute a contacting/separating mechanism (a drum unit pivoting mechanism) together with a pivot lever 83 (described later) in the main casing 2. The drum unit pivoting mechanism pivots the process frame 12 between an initial position and a pivot position. The protruding parts 75 are formed in a columnar shape and are disposed near the upper rear edges of the side plates 52. The protruding parts 75 protrude outward from the rear end of the process unit 12 in the axial direction of the photosensitive drums 16.

The rail parts 76 are formed continuously with the bottom edges of the side plates 52, protruding inward in the left or right direction. The front end of each rail part 76 is bent so as to slope upward toward the front.

Cutout parts 77 are formed in the left side plate 52L at positions corresponding to the developer guide grooves 32 of the left side walls 31L, and cutout parts 78 are formed in the right side plate 52R at positions corresponding to the electrodes 39 provided on the right side walls 31R.

The cutout parts 77 formed in the left side plate 52L are U-shaped in a side view and are open on the top. The shape of the cutout parts 77 conforms to the peripheral surfaces of the developing rollers 20. The cutout parts 78 formed in the right side plate 52R are substantially rectangular-shaped in a side view with a front-to-rear dimension long enough to expose all electrodes 39 on the respective right side walls 31R when viewed from the side.

When mounting the process cartridges 13 in the process frame 12, first the yellow process cartridge 13Y, magenta process cartridge 13M, and cyan process cartridge 13C, i.e., the non-black process cartridges 13, are mounted in the color mounting portion 63 of the cartridge support frame 51.

The operator mounts the non-black process cartridges 13 in the color mounting portion 63 by gripping the grip part 40 of each process cartridge 13, positioning the process cartridge 13 so that the front and/or rear edges of the process cartridge 13 are flush with the partitioning plates 59 while the drum couplings 35 are aligned with the drum support grooves 56, and inserts the process cartridge 13 downward into the color mounting portion 63 so that the drum couplings 35 are fitted into the drum support grooves 56.

At this time, the process cartridge 13 is positioned in the color mounting portion 63 so that the bottom edges of both side walls 31 of the process cartridge 13 contact the cartridge support rails 65 on both frame side walls 53. After sequentially mounting each non-black process cartridge 13 in the color mounting portion 63, the left and right side plates 52 are assembled on the process frame 12.

The operator assembles the left and right side plates 52 on the process frame 12 by positioning each side plate 52 relative to the cartridge support frame 51 so that the drum couplings 35 or drum support members (not shown) on the non-black process cartridges 13 are fitted into the drum retaining holes 73 formed in the side plates 52 and so that the drum support grooves 56 formed in the cartridge support frame 51 are aligned with the drum retaining holes 73 and the black support hole 71 formed in each side plate 52.

Consequently, the non-black process cartridges 13 are non-detachably supported in the process frame 12, as shown in FIG. 4. Accordingly, the photosensitive drums 16 provided in the non-black process cartridges 13 are non-detachably supported between the side plates 52 and capable of rotating relative to the side plates 52. That is, the photosensitive drums 16 provided in the non-black process cartridges 13 are fixed in the process frame 12.

Further, the stoppers 33 provided on each process cartridge 13 are inserted into the corresponding stopper fitting holes 74 in a direction from the inside of the side plate 52 toward the outside. Further, the cutout parts 77 formed in the left side plate 52L are aligned with the developer couplings 43 of the process cartridges 13 in the left-to-right direction, while the cutout parts 78 formed in the right side plate 52R are aligned with the electrodes 39 on the process cartridges 13 in the left-to-right direction. Accordingly, the developer couplings 43 and electrodes 39 are exposed on the left and right sides of the left side plate 52L and right side plate 52R, respectively.

11

Next, the black process cartridge **13K** is mounted in the black mounting portion **62**. As shown in FIG. 5, the black process cartridge **13K** is mounted in the black mounting portion **62** in a manner similar to that used for mounting the non-black process cartridges **13** in the color mounting portion **63**. Specifically, the operator grips the grip part **40** of the black process cartridge **13K** and positions the black process cartridge **13K** so that the front end of the black process cartridge **13K** is aligned with the front beam **54** and the rear end is aligned with the black partitioning plate **61**, and so that the drum couplings **35** are aligned with the black support holes **71** and the stoppers **33** with the stopper fitting holes **74** (see FIG. 6). Next, the operator inserts the black process cartridge **13K** downward into the black mounting portion **62**, fitting the drum couplings **35** into the black support holes **71** and the stoppers **33** into the stopper fitting holes **74**.

Since the black support holes **71** and stopper fitting holes **74** are open on the top, the black process cartridge **13K** is detachably supported in the process frame **12**. Consequently, the photosensitive drum **16** provided in the black process cartridge **13K** is detachably supported between the side plates **52** and is capable of rotating relative to the side plates **52**. In other words, the photosensitive drum **16** of the black process cartridge **13K** is detachably provided in the process frame **12**. At this time, mounting of all process cartridges **13** in the process frame **12** is completed.

As shown in FIGS. 7 and 8, the main casing **2** includes a pair of left and right casing side walls **81** disposed parallel to each other on either side of the conveying belt **24**, a reference shaft **82** spanning between the casing side walls **81**, the pivot lever **83** spanning between the casing side walls **81**, pressing members **103**, and a translation cam mechanism **84** disposed on the left side of the left casing side wall **81**. Hereinafter the casing side wall **81** on the left side will be referred to as the left casing side wall **81L** and the casing side wall **81** on the right side the right casing side wall **81R**. The reference shaft **82** adjusts a position of the drum unit in the main casing **2**.

As shown in FIG. 8, each of the casing side walls **81** includes a frame support part **85**, and a black positioning plate **86**. A frame guide groove **87** is also formed in each casing side wall **81**. The frame support parts **85** are provided along the lower edge of both casing side walls **81** in the front-to-rear direction and protrude inward from the inner surfaces of the casing side walls **81** in the respective left and right directions. A protrusion **88** is formed on the front end of each frame support part **85**. The protrusion **88** has an arc shape in a side cross-sectional view and protrudes upward.

Each black positioning plate **86** is a flat plate that is substantially rectangular in a side view. The black positioning plate **86** is disposed rearward of the corresponding protrusion **88** and protrudes upward from the top surface of the corresponding frame support part **85**. A portion is cut out from the top edge of each black positioning plate **86** to form a shallow V-shape in a side view so that the top edge of the black positioning plate **86** is lower in the center than the front and rear edges. The black positioning plates **86** are disposed in positions for receiving the black support parts **72** of the side plates **52** when the process frame **12** is mounted in the main casing **2** (see FIG. 6).

The frame guide grooves **87** extend in the front-to-rear direction through substantially the vertical center of the casing side walls **81** and are shaped for receiving the protruding parts **75** provided on the process frame **12**. A step part **108** formed substantially in a crank-shape when viewed from the side is formed in the front end of each frame guide

12

groove **87**. A pivot guide groove **89** that is wider vertically than the frame guide groove **87** is formed in the rear end of each frame guide groove **87**.

The pivot guide groove **89** has a rectangular shape in a side view. As will be described later in greater detail, the pivot guide grooves **89** function to receive the protruding parts **75** when the process frame **12** is mounted in the main casing **2** and to guide vertical movement of the protruding parts **75** when the process frame **12** is pivoted, thereby restricting the pivoting range of the process frame **12**.

As shown in FIGS. 7 and 8, through-holes **90** are formed in both casing side walls **81** at positions corresponding to the developer couplings **43** or electrodes **39**. The through-holes **90** formed in the left casing side wall **81L** are formed in a circular shape in a side view that corresponds to the shape of the developer couplings **43**, while the through-holes **90** formed in the right casing side wall **81R** are rectangular-shaped in a side view and are large enough to expose all electrodes **39**.

As shown in FIG. 8, the reference shaft **82** is formed of metal or the like in a rod shape and extends between the lower rear ends of the casing side walls **81**. The bottom surface on the rear end of the process frame **12** contacts the reference shaft **82** when the process frame **12** is completely mounted in the main casing **2**.

As shown in FIG. 7, the pivot lever **83** constitutes a contacting/separating mechanism together with the protruding parts **75** of the process frame **12**. The pivot lever **83** is pivotable in a plane perpendicular to the axial directions of photosensitive drums **16**. The pivot lever **83** includes a pivot shaft **91**, and a pair of lever members **92** coupled one with each end of the pivot shaft **91** so as to be incapable of rotating relative to the pivot shaft **91**.

As shown in FIG. 8, the pivot shaft **91** has a rod shape and is disposed in substantially the vertical center of the casing side walls **81** toward the rear ends thereof. Both ends of the pivot shaft **91** are rotatably inserted through through-holes **93** penetrating the casing side walls **81** and protrude outward from the casing side walls **81** in the left and right directions.

The lever members **92** are formed in arm-like shapes. The rear ends of the lever members **92** are coupled with the pivot shaft **91** at a point outside the casing side walls **81** relative to the left and right directions. Through this construction, the front ends of the lever members **92** pivot vertically, i.e., in a direction orthogonal to the left-to-right direction, about the pivot shaft **91**.

As shown in FIG. 8, the pressing member **103** is disposed on the inner side of each casing side wall **81**. Each pressing member **103** includes a rib **104** supported on the main casing **2**, a pressing protrusion **105** capable of sliding vertically, and a compression spring **106** disposed between the rib **104** and pressing protrusion **105**.

The rib **104** is disposed above and slightly forward of the black positioning plate **86** and is fixed in position relative to the main casing **2** by a support frame (not shown). The pressing protrusion **105** is disposed directly below the rib **104** while separated therefrom. The pressing protrusion **105** is positioned to contact and apply pressure to the tops of the protrusions **34** formed on the process cartridge **13** when the process frame **12** is mounted in the main casing **2**.

As shown in FIG. 7, the translation cam mechanism **84** is disposed on the outer surface of the left casing side wall **81L**. The translation cam mechanism **84** selectively transmits drive power to each of photosensitive drums **16**. The translation cam mechanism **84** includes a translation cam **94**

13

serving as a switching member, and four drive transmission gear parts **95** for transmitting a drive force to each of the photosensitive drums **16**.

The translation cam **94** is formed from a flat plate having a substantially rectangular shape in a side view. While the function of the translation cam **94** will be described later in greater detail, the translation cam **94** is capable of moving in the front-to-rear direction.

A protruding part **107** is formed on the top edge of the translation cam **94** near the rear end thereof. The protruding part **107** is substantially triangular in a side view and protrudes upward from the top edge of the translation cam **94**. In other words, the protruding part **107** protrudes toward the pivot lever **83**. The top of the protruding part **107** is formed level. The protruding part **107** is disposed so as to push the left lever member **92** of the pivot lever **83** upward when the top end of the protruding part **107** contacts this lever member **92**.

As shown in FIG. 12, an elongated hole is formed in the translation cam **94**. The hole is elongated in the front-to-rear direction and has a front-to-rear length nearly equivalent to the front-to-rear length of the process frame **12**. A cylindrical part **96** is integrally formed with the translation cam **94** around the elongated hole, protruding leftward from the peripheral edge of the elongated hole.

As shown in FIG. 13, the cylindrical part **96** is integrally formed with four retracting parts **97** corresponding to the four drive transmission gear parts **95** that protrude leftward from the left edge of the cylindrical part **96**, and sloped parts **98** formed continuously with the rear edges of the retracting parts **97**. The front-to-rear dimension of the forwardmost retracting part **97** (i.e., the retracting part **97** corresponding to the black process cartridge **13K**) is shorter than the same dimension of the other three retracting parts **97**. The protruding length of each of the sloped parts **98** grows gradually shorter in the direction rearward from the rear edge of the corresponding retracting parts **97**.

Each drive transmission gear part **95** includes a gear **99**, a shaft **100**, and a male drum coupling member **101**. A drive force generated by a motor (not shown) provided in the main casing **2** is inputted into the gears **99**. The shaft **100** is integrally formed with the corresponding gear **99** and extends rightward from the same.

The right end of the male drum coupling member **101** is formed to mate with the female drum coupling member **37** of drum coupling **35**, while the left end is formed in a cylindrical shape for receiving the shaft **100** so that the shaft **100** cannot rotate relative to the male drum coupling member **101**. A bridge part **102** having a diameter longer than the vertical length of the cylindrical part **96** is formed on the left end of the male drum coupling member **101**.

A compression spring (not shown) is also mounted between the gear **99** and the bridge part **102** for urging the male drum coupling member **101** rightward.

Next, an operation for mounting the process frame **12** in the main casing **2** will be described with referred to FIGS. 9-12. When mounting the process frame **12** in the main casing **2**, the operator first grips the first frame handle **57** and second frame handle **58** on the process frame **12** and inserts the protruding parts **75** on the process frame **12** into the front ends of the frame guide grooves **87** formed in the main casing **2**. While gripping the first frame handle **57**, the operator pushes the process frame **12** rearward.

At this point, the protruding parts **75** are guided along the frame guide grooves **87** and contact the step parts **108** formed in the frame guide grooves **87**, as shown in FIG. 9. In this state, the process frame **12** is oriented along a slant,

14

with the rear end positioned lower than the front end, and the rail parts **76** along the lower edges of the process frame **12** are positioned outside the main casing **2**.

Next, while continuing to grip the first frame handle **57**, the operator grips the second frame handle **58** and lifts the second frame handle **58** upward. As a consequence, the protruding parts **75** move upward in the step parts **108** of the frame guide grooves **87**, as shown in FIG. 10. When the process frame **12** is brought to a level state, the bottom edge of the process frame **12** on the rear end thereof is inserted into the main casing **2**.

At this time, the rail parts **76** on the process frame **12** are in contact with the protrusions **88** of the main casing **2** so that the protrusions **88** support the rear ends of the process frame **12** on the bottom thereof, as shown in FIG. 11.

Next, the operator continues to push the process frame **12** rearward, while the protruding parts **75** are guided rearward along the frame guide grooves **87**. When the protruding parts **75** arrive in the pivot guide grooves **89** formed at the rear ends of the frame guide grooves **87**, the operation for mounting the process frame **12** is complete, as shown in FIG. 12.

Next, an operation of process unit **9** will be described with referred to FIGS. 12-21. The printer **1** includes a control unit (not shown) that sets the printer **1** to one among the monochrome mode, the color mode, and the drum detachable mounting mode. Details of each mode will be described below.

When the process frame **12** is completely mounted in the main casing **2** as shown in FIG. 12, the protruding parts **75** are positioned on the lower edges of the pivot guide grooves **89** and protrude outward from the casing side walls **81** in the respective left and right directions. Also at this time, the translation cam **94** is in a first position, which is a drum detachable mounting position, and the protruding part **107** is positioned on the rear side of the left lever member **92**. When the process frame **12** is completely mounted in the main casing **2**, the control unit (not shown) sets the printer **1** to the drum detachable mounting mode.

Further, the drum couplings **35** are in confrontation with the cylindrical part **96** of the translation cam **94** in the left-to-right direction so that the drum couplings **35** are exposed through the cylindrical part **96** on the left side of the main casing **2**.

Similarly, the developer couplings **43** are in confrontation with the through-holes **90** formed in the left casing side wall **81L** in the left-to-right direction, whereby the developer couplings **43** are exposed through the through-holes **90** on the left side of the main casing **2**. By inserting male coupling members (not shown) into the through-holes **90** to be coupled with the female developer coupling members **44**, a drive force can be transmitted to the developer couplings **43**.

Similarly, the electrodes **39** are in confrontation with the through-holes **90** formed in the right casing side wall **81R** in the left-to-right direction, whereby the electrodes **39** are exposed through the through-holes **90** on the right side of the main casing **2**.

Further, as shown in FIG. 13, the bridge parts **102** of all the male drum coupling members **101** are in contact with the corresponding retracting parts **97** of the translation cam **94**, so that all male drum coupling members **101** are retracted leftward. In other words, the translation cam **94** has uncoupled the female drum coupling members **37** (FIG. 14) of all photosensitive drums **16** from all male drum coupling members **101**.

Further, the rail parts **76** are no longer in contact with the protrusions **88**, as shown in FIG. 14. Consequently, the black

15

support parts 72 are in contact with and supported by the black positioning plates 86. The rear end of the process frame 12 also contacts the reference shaft 82 and is supported thereon. With this configuration, the process frame 12 is disposed in a level orientation in the main casing 2.

As shown in FIG. 15(a), all of the photosensitive drums 16 are in contact with the conveying belt 24. As shown in FIG. 15(b), the pressing protrusions 105 of the pressing members 103 are in contact with the tops of the protrusions 34, applying a downward pressure to the protrusions 34. More specifically, the pressing members 103 press both side walls 31 and the drum partition 41 of the black process cartridge 13K downward toward a position between the corresponding photosensitive drum 16 and stopper 33.

Further, the rail parts 76 are disposed in confrontation with but are separated from the frame support parts 85 vertically. Consequently, the printer 1 is in the drum detachable mounting mode in which the photosensitive drums 16 can be detached from the main casing 2.

When the control unit (not shown) sets the printer 1 to the monochrome mode, the translation cam 94 is subsequently moved forward as shown in FIG. 16, and then the protruding part 107 contacts the left lever member 92 and the lever member 92 pivots upward along the front sloped surface of the protruding part 107. The right lever member 92 consequently pivots upward together with the left lever member 92. As the lever members 92 pivot, the upper edges of the lever members 92 contact the bottoms of the protruding parts 75 and lift the protruding parts 75 upward. In other words, the lever member 92 pivots from the initial position shown in FIG. 12 to the pivot position shown in FIG. 16.

At the same time, the bridge part 102 of the male drum coupling member 101 corresponding to the black process cartridge 13K is released from its contact with the retracting part 97 of the cylindrical part 96 and advances rightward along the sloped part 98, as shown in FIG. 17.

When the translation cam 94 reaches a second position, which is a monochrome image-forming position, the front end of the lever member 92 is in contact with the top of the protruding part 107, as shown in FIG. 16. At this point, the lever members 92 have lifted the protruding parts 75 to the top ends of the pivot guide grooves 89.

Through this operation, the process frame 12 is pivoted about the central axis of the photosensitive drum 16 in the black process cartridge 13K, with the rear end of the process frame 12 rising upward, as illustrated in FIGS. 18 and 19(b).

Further, the photosensitive drum 16 of the black process cartridge 13K is in contact with the conveying belt 24, while the photosensitive drums 16 of the non-black process cartridges 13 confront the conveying belt 24 vertically but are separated therefrom, as shown in FIG. 19(a).

At the same time, the male drum coupling member 101 corresponding to the black process cartridge 13K is coupled with the female drum coupling member 37 of the black process cartridge 13K so as to share the same central axis, as shown in FIG. 17. The male drum coupling members 101 corresponding to the non-black process cartridges 13 are still retracted leftward and are therefore not coupled with the corresponding female drum coupling members 37.

In this state, the printer 1 is in the monochrome mode for forming monochromatic images. When a drive force is inputted into each of the gears 99 in this monochrome mode, the drive force is transmitted to the black photosensitive drum 16, but not to the non-black photosensitive drums 16, enabling the formation of images in black only. At this time, the male drum coupling members 101 corresponding to the

16

non-black photosensitive drums 16 rotate while remaining disengaged from the photosensitive drums 16.

When the control unit (not shown) sets the printer 1 to the color mode, the translation cam 94 is moved farther forward as shown in FIG. 20, and then the lever member 92 pivots downward along the sloped rear surface of the protruding part 107, and consequently both lever members 92 pivot downward. As the lever members 92 pivot downward, the protruding parts 75 also drop downward.

At the same time, as shown in FIG. 21, the bridge parts 102 of all male drum coupling members 101 are disengaged from the retracting parts 97 of the cylindrical part 96 and advance rightward along the sloped parts 98.

When the translation cam 94 reaches a third position, which is the color image-forming position, the front end of the left lever member 92 no longer contacts the protruding part 107, as shown in FIG. 20. At this time, the protruding parts 75 are once again disposed in the bottom part of the pivot guide grooves 89. Accordingly, as shown in FIG. 22, the rear end of the process frame 12 again contacts the reference shaft 82 so that the process frame 12 is in a level orientation. In addition, as in the drum detachable mode, all of the photosensitive drums 16 are in contact with the conveying belt 24.

Further, all of the male drum coupling members 101 are coupled with the corresponding female drum coupling members 37 and share the central axis of the female drum coupling members 37, as shown in FIG. 21. Thus, the printer 1 is in a color mode for forming color images. When a drive force is inputted into the gears 99 in the color mode, the drive force is transmitted to all photosensitive drums 16 in order to form a color image.

As shown in FIG. 5, the color laser printer 1 is provided with four photosensitive drums 16. The photosensitive drums 16 provided in the non-black process cartridges 13 are fixed to the process frame 12, while the photosensitive drum 16 provided in the black process cartridge 13K is detachably provided in the process frame 12.

Accordingly, the black photosensitive drum 16 can be removed and replaced while the non-black photosensitive drums 16 remain fixed to the process frame 12. As a result, when the non-black photosensitive drums 16 are used with low frequency while the black photosensitive drum 16 is used with high frequency, this construction allows the user to efficiently maintain the frequently used black photosensitive drum 16 while preserving the relative positioning of the infrequently used non-black photosensitive drums 16. Hence, the printer 1 allows efficient maintenance of the photosensitive drums 16 based on usage frequency while ensuring high-quality image formation.

In the printer 1, as shown in FIG. 5, the non-black photosensitive drums 16 are non-detachably supported between the side plates 52 of the process frame 12. Accordingly, the pitch between neighboring non-black photosensitive drums 16 can be preserved with greater precision.

Further, in the printer 1, as shown in FIG. 5, the black photosensitive drum 16 is detachably supported between the side plates 52 of the process frame 12. Accordingly, the black photosensitive drum 16 can be removed from the process frame 12 while preserving the pitch between neighboring photosensitive drums 16 with precision.

In the printer 1, the black photosensitive drum 16 is disposed in the front side of the process frame 12, as shown in FIG. 1. Hence, when the user is mounting and removing the process frame 12 on the front side, the black photosensitive drum 16 is positioned nearest the user. This construc-

17

tion makes it easy to remove the black photosensitive drum 16 from the process frame 12, facilitating replacement operations.

As illustrated in FIG. 19, the black photosensitive drum 16 is provided in the front end of the process frame 12 according to the printer 1, and the non-black photosensitive drums 16 can be separated from the conveying belt 24 by pivoting the rear end of the process frame 12 upward about the axis of the black photosensitive drum 16.

With this configuration, all non-black photosensitive drums 16 provided in the process frame 12 can be separated from the conveying belt 24 as a unit, while preserving the relative positioning of all photosensitive drums 16. In other words, the non-black photosensitive drums 16 can be separated from the conveying belt 24 while maintaining the pitch between neighboring photosensitive drums 16. Therefore, this construction ensures high-quality image formation while suitably reducing wear on the photosensitive drums 16.

In the printer 1 according to the preferred embodiment, the process frame 12 can slide in the front-to-rear direction. In this way, the four photosensitive drums 16 can be operated as a unit.

With the printer 1, the pivot lever 83 and protruding parts 75 (contacting/separating mechanism) for pivoting the process frame 12 are disposed on the rear side of the process frame 12, as shown in FIG. 14. Hence, this construction makes it easy to apply the contacting/separating mechanism to the rear end of the process frame 12. Consequently, the process frame 12 can easily be pivoted.

More specifically, as illustrated in FIGS. 19(a) and 19(b), the printer 1 has the pivot lever 83 that can pivot in a direction orthogonal to the left-to-right direction, and protruding parts 75 protruding outward from the rear end of the process frame 12 in the left and right directions. The process frame 12 can be pivoted when the pivot lever 83 contacts and moves the protruding part 75. Hence, the process frame 12 can be pivoted through a simple contacting/separating mechanism.

Further, when all photosensitive drums 16 are in contact with the conveying belt 24, as illustrated in FIGS. 15(a) and 15(b), the rear end of the process frame 12 is in contact with the reference shaft 82, thereby positioning the process frame 12 relative to the conveying belt 24. Accordingly, the photosensitive drums 16 can be positioned relative to the conveying belt 24 in order to reliably maintain suitable contact between the photosensitive drums 16 and the conveying belt 24.

Further, the pressing members 103 press downward on the arc-shaped protrusions 34 formed on the side walls 31 of the black process cartridge 13K, as shown in FIG. 15. Further, the protrusions 34 share a central axis with the photosensitive drums 16, as shown in FIG. 6. Accordingly, when pressed against the protrusions 34, the pressing members 103 constantly apply a force to the black photosensitive drum 16 toward the central axis thereof. This force is constantly applied toward the black photosensitive drum 16, even when the position of contact between the pressing members 103 and protrusions 34 changes as the process frame 12 is pivoted.

More specifically, as shown in FIG. 15, the pressing members 103 press the side walls 31 toward a position between the black photosensitive drum 16 and the corresponding stoppers 33 separated from the black photosensitive drum 16. Accordingly, the pressing members 103 can apply pressure to both the photosensitive drum 16 and the

18

stoppers 33. Consequently, the black photosensitive drum 16 is always fixed in position, even when the process frame 12 is pivoted.

Further, in the printer 1, each of the male drum coupling members 101 is coaxially coupled with a corresponding photosensitive drum 16 when in the color mode. By sharing the same axis as the corresponding photosensitive drums 16, the male drum coupling members 101 can input a drive force to the photosensitive drums 16. As a result, a drive force can be precisely transmitted from the male drum coupling members 101 to the photosensitive drums 16.

The male drum coupling members 101 are uncoupled from the non-black photosensitive drums 16 in the monochrome mode when the non-black photosensitive drums 16 are separated from the conveying belt 24. Hence, when the male drum coupling members 101 are uncoupled from the non-black photosensitive drums 16, the rotation of the non-black photosensitive drums 16 is halted when the photosensitive drums 16 are separated from the conveying belt 24, even though the male drum coupling members 101 continue to rotate. Accordingly, the translation cam mechanism 84 can be provided in a simple structure for coupling or uncoupling the male drum coupling members 101 from the photosensitive drums 16.

The printer 1 also has a drum detachable mounting mode in which the photosensitive drums 16 is allowed to be mounted on or separated from the main casing 2. Accordingly, the process frame 12 having the photosensitive drums 16 can be removed in the drum detachable mounting mode, which is different from the modes for forming images (the color mode and monochrome mode). As a result, the process frame 12 can be removed while image-forming operations are not being performed.

With the printer 1, the translation cam 94 provided in the translation cam mechanism 84 is selectively movable to one of the first position corresponding to the drum detachable mounting mode, the second position corresponding to the monochrome mode, and the third position corresponding to the color mode. Hence, the modes of the printer 1 can be switched through a simple operation of sliding the translation cam 94.

By placing the translation cam 94 in the first position, all male drum coupling members 101 can be uncoupled from all photosensitive drums 16, while all photosensitive drums 16 are placed in contact with the conveying belt 24. Through this operation, the process frame 12 can be removed while the rotation of all photosensitive drums 16 is halted.

By placing the translation cam 94 in the second position, the male drum coupling member 101 corresponding to the black photosensitive drum 16 is coupled to this photosensitive drum 16, while the other male drum coupling members 101 are uncoupled from the non-black photosensitive drums 16, and the black photosensitive drum 16 is placed in contact with the conveying belt 24 while the non-black photosensitive drums 16 are separated from the conveying belt 24. Accordingly, the black photosensitive drum 16 in contact with the conveying belt 24 can be used for image formation, while the rotation of the non-black photosensitive drums 16 separated from the conveying belt 24 is halted.

By placing the translation cam 94 in the third position, all male drum coupling members 101 are coupled with all corresponding photosensitive drums 16, and all of the photosensitive drums 16 are placed in contact with the conveying belt 24. In this mode, all photosensitive drums 16 can be used for image formation.

In other words, movement of the translation cam 94 among the first, second, and third positions associates opera-

19

tions of the translation cam **94** for coupling or uncoupling the male drum coupling members **101** from corresponding photosensitive drums **16** with the operations of the contacting/separating mechanism (the pivot lever **83** and protruding parts **75**) for placing the photosensitive drums **16** in contact with or separating the photosensitive drums **16** from the conveying belt **24**. Hence, through a simple construction, the translation cam **94** can associate operations of the translation cam mechanism **84** with operations of the contacting/separating mechanism.

Further, when the translation cam **94** is placed in the first position, the top of the protruding part **107** is out of contact in the pivot lever **83** and, hence, the pivot lever **83** is not pivoted upward. Consequently, the photosensitive drums **16** are not separated from the conveying belt **24**.

When the translation cam **94** is placed in the second position, the top of the protruding part **107** is contacting the pivot lever **83**, which contact applies a force to the pivot lever **83** that pivots the pivot lever **83** upward. At this time, the non-black photosensitive drums **16** are separated from the conveying belt **24**.

When the translation cam **94** is placed in the third position, the top of the protruding part **107** no longer contacts the pivot lever **83**, as in the first position. Hence, the pivot lever **83** is not pivoted upward and the photosensitive drums **16** are not separated from the conveying belt **24**.

In other words, switching the translation cam **94** between the first, second, and third positions determines whether the top of the protruding part **107** in the translation cam **94** is contacting or not contacting the pivot lever **83**. The pivot lever **83** is pivoted upward when the top of the protruding part **107** contacts the pivot lever **83**. The non-black photosensitive drums **16** are separated from the conveying belt **24** only when the pivot lever **83** is pivoted upward. As a result, the position of the translation cam **94** determines whether the non-black photosensitive drums **16** are in contact with or separated from the conveying belt **24**.

What is claimed is:

1. An image forming apparatus comprising:

a main casing;

a process frame movable into and out of the main casing in a front-to-rear direction;

and a plurality of process cartridges supported by the process frame, the plurality of process cartridges being movable together with the process frame relative to the main casing in a state where the plurality of process cartridges is supported by the process frame,

one of the plurality of process cartridges including:

a drum cartridge movable relative to the process frame in a gravitational direction, the drum cartridge including a photosensitive drum rotatable about a drum axis extending in a left-right direction, the left-right direction crossing both of the front-to-rear direction and the gravitational direction; and

a toner cartridge configured to accommodate toner to be supplied to the photosensitive drum, the toner cartridge being removable relative to the process frame in the gravitational direction.

20

2. The image forming apparatus according to claim 1, wherein the process frame includes a drum supporting groove configured to support the drum cartridge.

3. The image forming apparatus according to claim 2, wherein the drum cartridge includes a stopper, and wherein the process frame includes a stopper support groove configured to support the stopper.

4. The image forming apparatus according to claim 3, wherein the stopper is a cylindrical rib protruding outward from a side wall of the drum cartridge in the left-right direction.

5. The image forming apparatus according to claim 4, wherein the stopper support groove is positioned forward of the drum supporting groove.

6. The image forming apparatus according to claim 2, wherein the process frame includes a drum-cartridge support rail configured to contact a side wall of the drum cartridge, the drum-cartridge support rail being positioned below and inward of the drum supporting groove in the left-right direction.

7. The image forming apparatus according to claim 1, wherein the drum cartridge includes a guide member configured to guide the toner cartridge.

8. The image forming apparatus according to claim 7, wherein the guide member includes a groove formed at an inner surface of a side wall of the drum cartridge in the left-right direction.

9. The image forming apparatus according to claim 1, wherein the toner cartridge is detachably supported by the drum cartridge.

10. The image forming apparatus according to claim 9, wherein the toner cartridge includes a developing roller and a developer coupling part configured to receive drive force for driving the developing roller; and wherein the drum cartridge has a developer coupling support groove configured to support the developer coupling part.

11. The image forming apparatus according to claim 10, wherein the process frame includes a cutout that is positioned corresponding to the developing roller in a state where the toner cartridge is supported by the process frame, the cutout being positioned forward of the drum supporting groove.

12. The image forming apparatus according to claim 1, wherein the drum cartridge includes a grip part at an upper surface of the drum cartridge.

13. The image forming apparatus according to claim 1, wherein the drum cartridge includes a scorotron charger configured to apply a charge of positive polarity to the photosensitive drum.

14. The image forming apparatus according to claim 1, wherein the photosensitive drum is configured to transfer a toner image formed on the photosensitive drum directly to a sheet.

* * * * *